

Don't Melt Your Cache: Low-Associativity with Heat-Sink

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Classical Problem: How to Manage a Cache

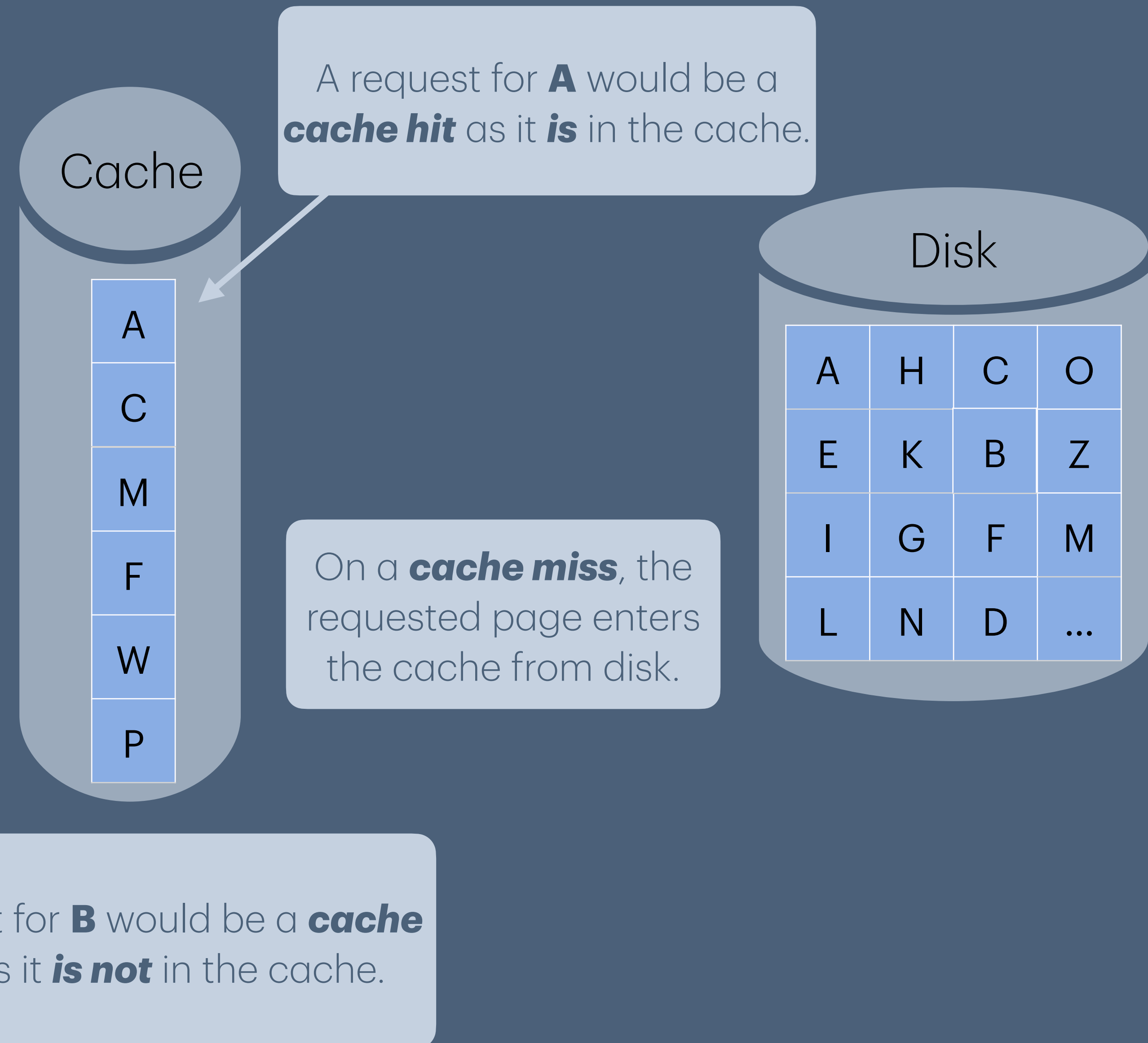
Given a sequence of page requests
 $\sigma = \langle \sigma_1, \sigma_2, \dots, \sigma_k \rangle$ and a cache of size n .

Cache hit: σ_i is in cache.

Cache miss: σ_i is not in cache.

The **replacement policy** determines what is **evicted** from cache on a cache miss.

Goal: Design a replacement policy to minimize **cache misses**.

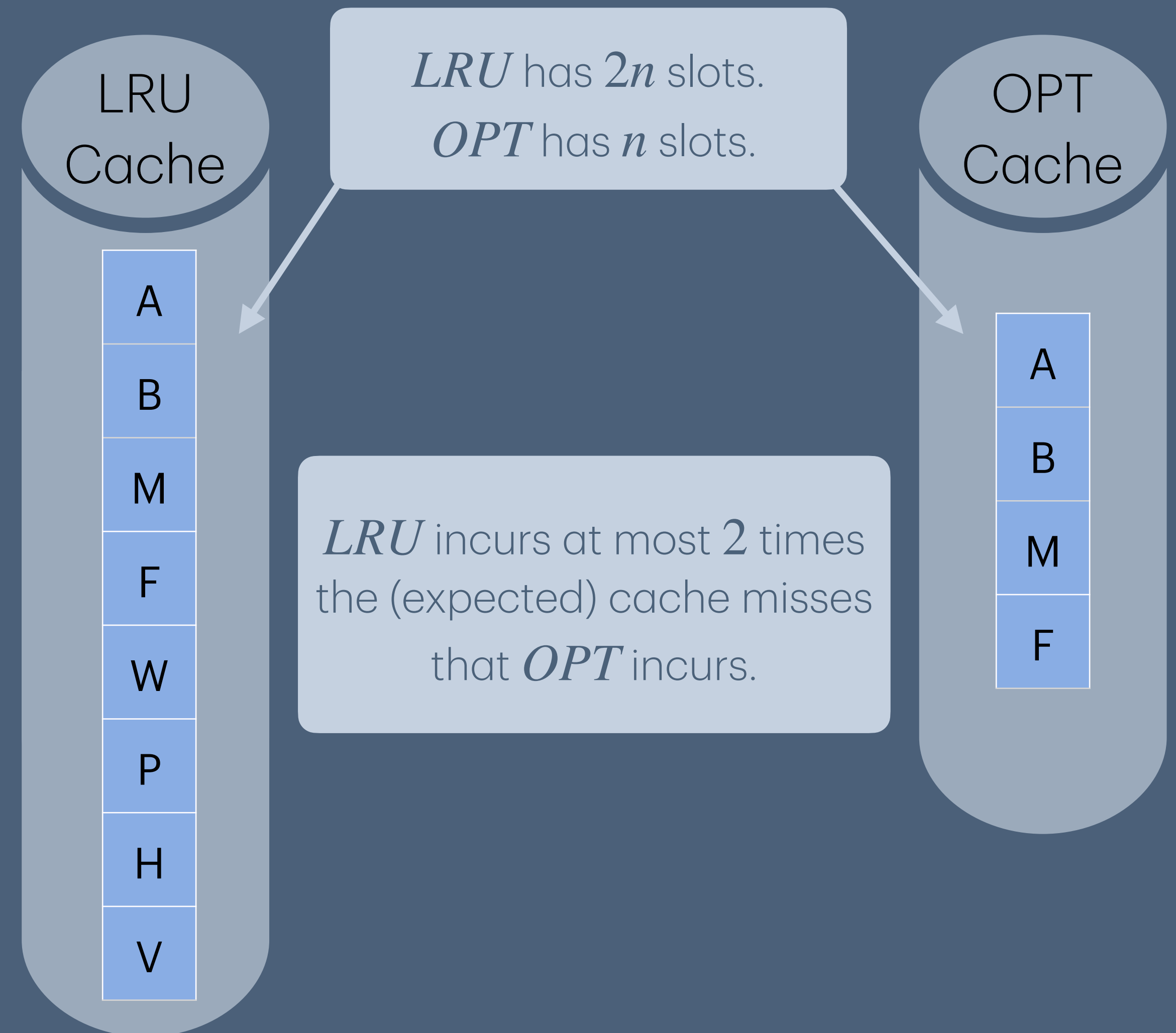


Seminal Result: *LRU* is 2-competitive

LRU: Evict the *Least-Recently-Used* page.

Thm [Sleator & Tarjan, 1985]: *LRU* is 2-competitive with 2-resource augmentation.

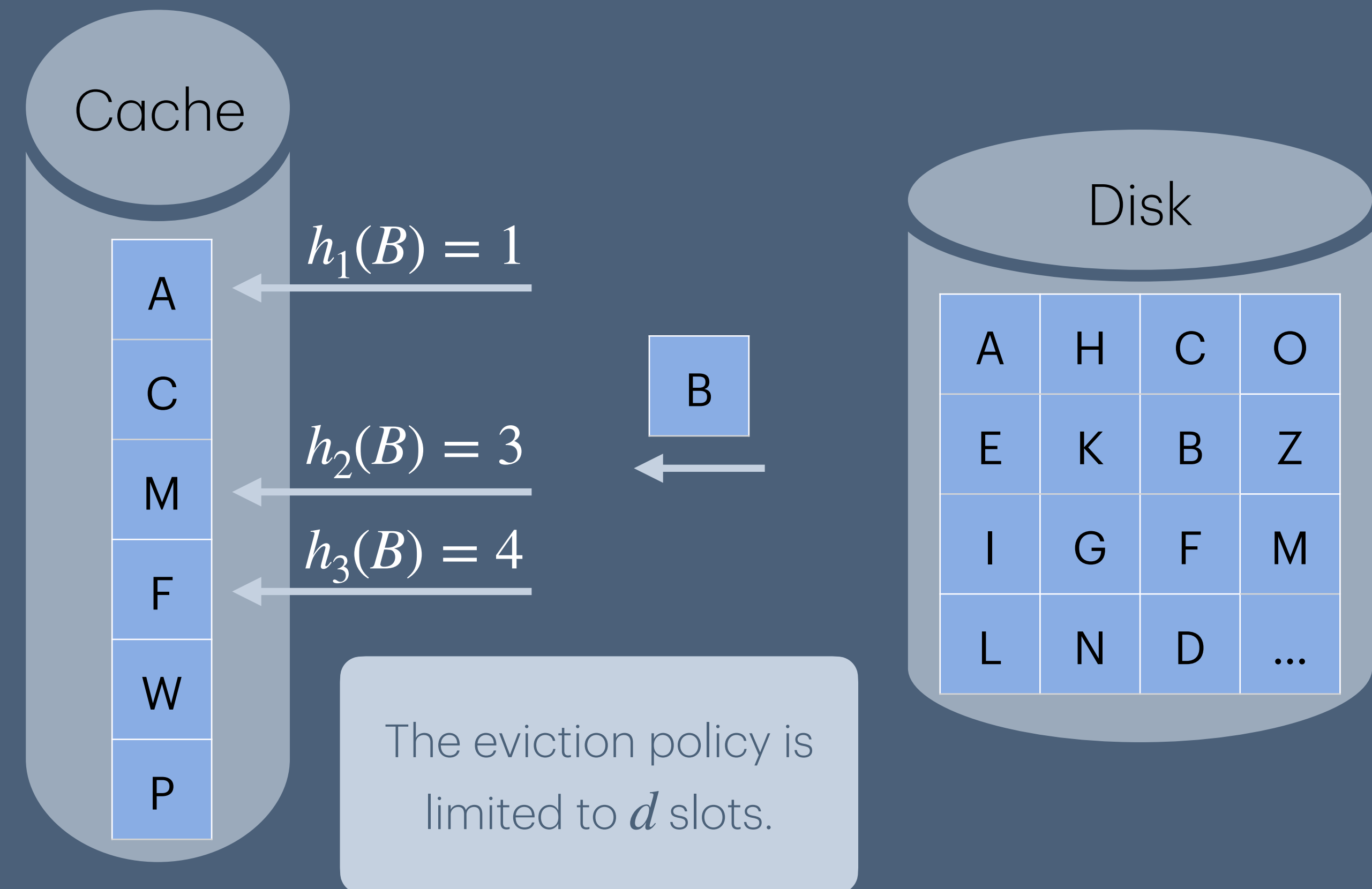
Surprisingly, this result *does not apply* to most real caches.



How to Manage a *Low-Associativity* Cache

***d*-associativity**: Each page x is restricted to a set of d slots in cache $H(x) = \{h_1(x), h_2(x), \dots, h_d(x)\}$.

Real caches have $d \leq 16$ to reduce cost and hardware complexity¹.



¹ [IEEE Micro'15, ISSCC'15, ISSCC'18, IEEE Micro'22]

Question: *Can a low-associativity policy be competitive with **OPT**?*

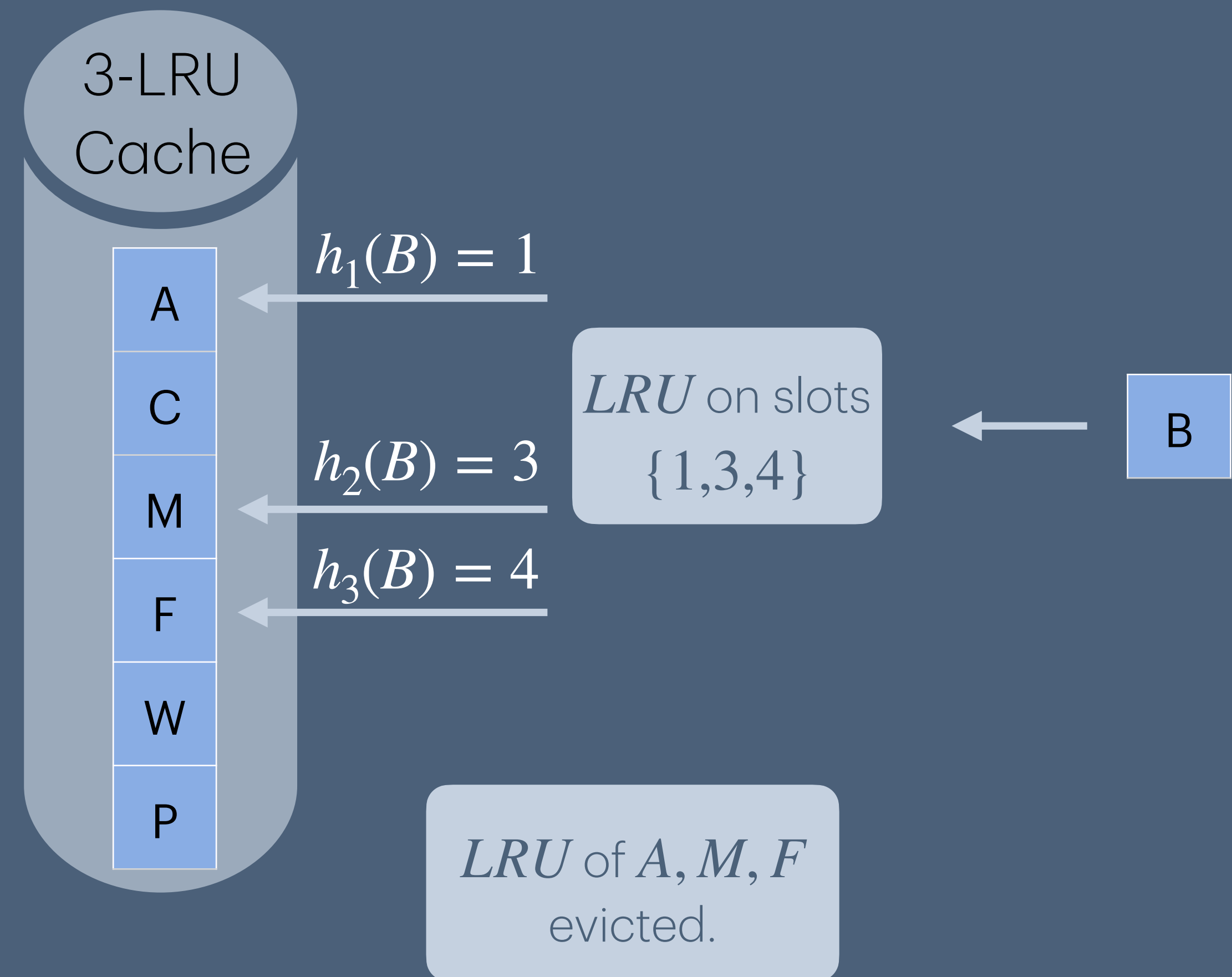
This is a lofty goal— we're comparing against **fully associative OPT**.

LRU with Restricted Associativity

d -*LRU*: perform *LRU* on the d uniformly random but fixed slots
 $H(x) = \{h_1(x), h_2(x), \dots, h_d(x)\}$.

d -*LRU* and variants are widely used in systems, without analytic bounds.¹

How does d -*LRU* compare with fully associative *OPT*?



¹ [ISCA'93, ISPASS'14, USENIX'20, IPDPS'20]

Results

d -LRU is *not* $O(1)$ -competitive with $O(1)$ augmentation for $d \in o(\log n / \log \log n)$.

We're comparing against **fully associative OPT**.

2-RAND is $O(1)$ -competitive with $O(1)$ augmentation.

Heat-Sink-LRU is $1 + o(1)$ -competitive with LRU when given $1 + o(1)$ augmentation for $d \in \omega(1)$.

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We'll come back to this result towards the end.

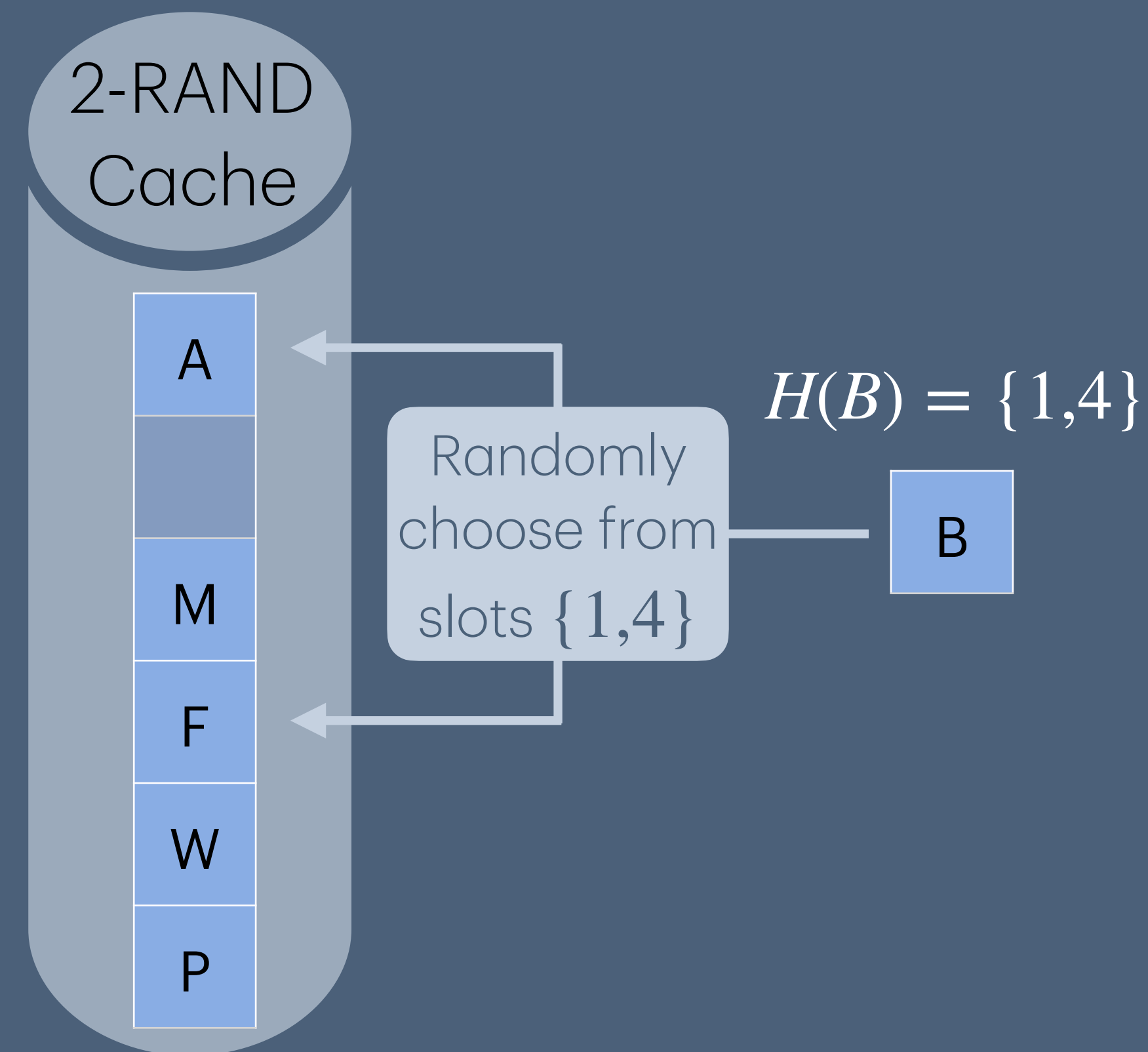
2-RAND: Competitive with $d = 2$

2-RAND: Randomly pick from the 2 uniformly random but fixed slots

$$H(x) = \{h_1(x), h_2(x)\}.$$

Thm: *2-RAND* is $O(1)$ -competitive with $O(1)$ augmentation.

Remarkably, random behavior is better than trying to do the “right thing.”

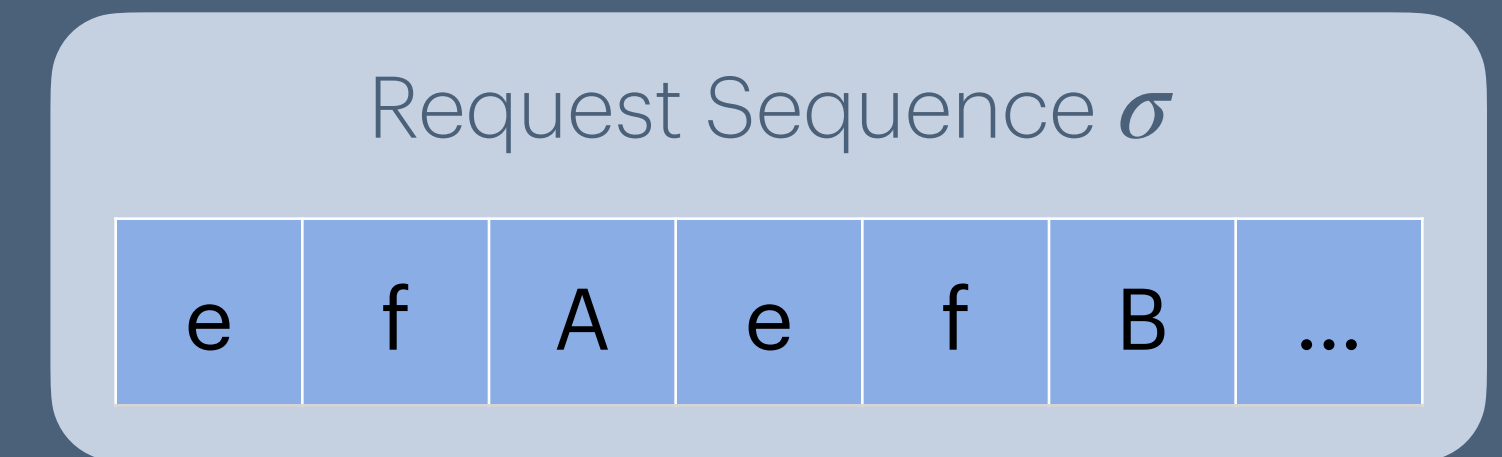
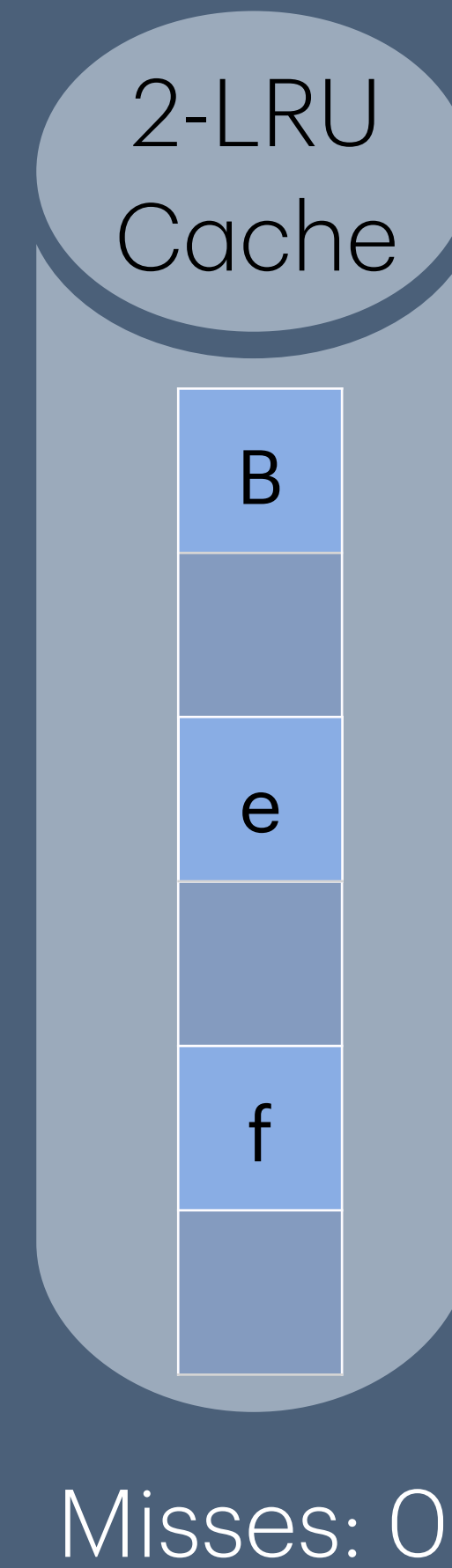
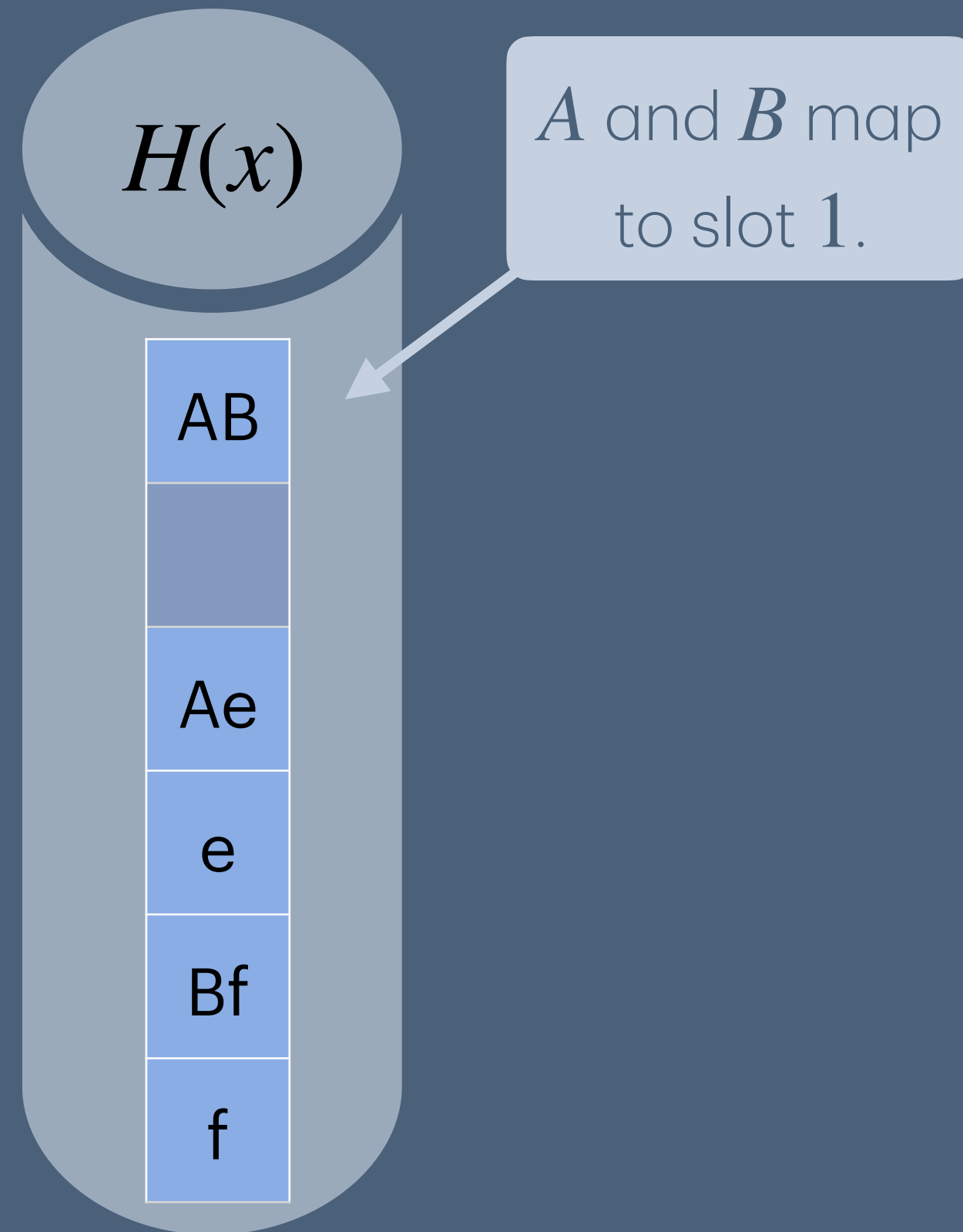


Question: Why is *d-LRU* not competitive while *2-RAND* is?

LRU: The Illusion of Choice

Consider the repeating sequence

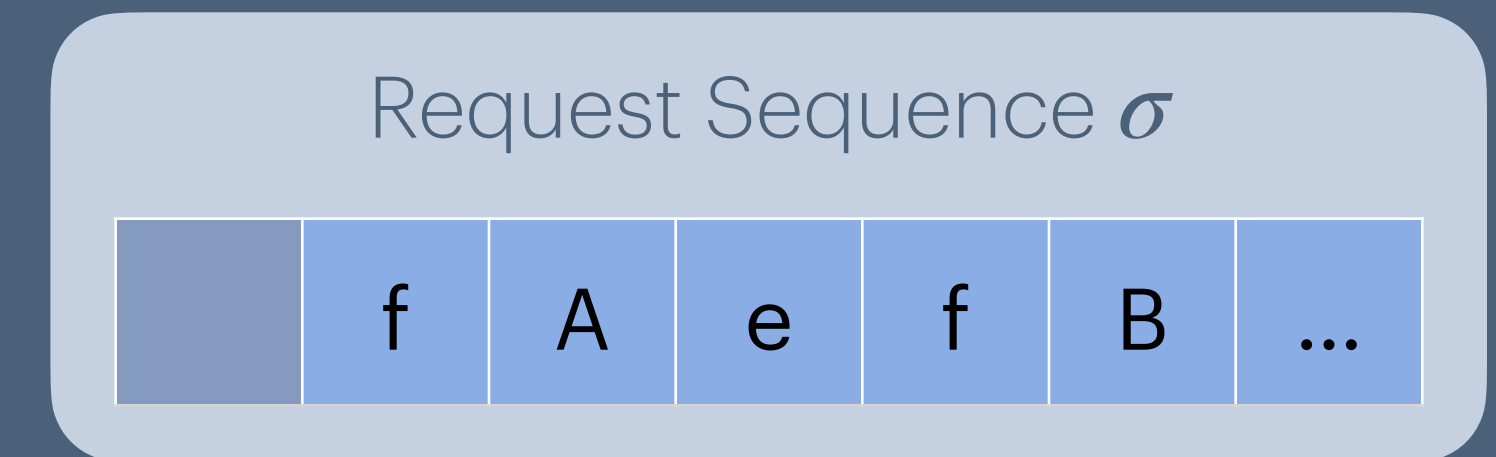
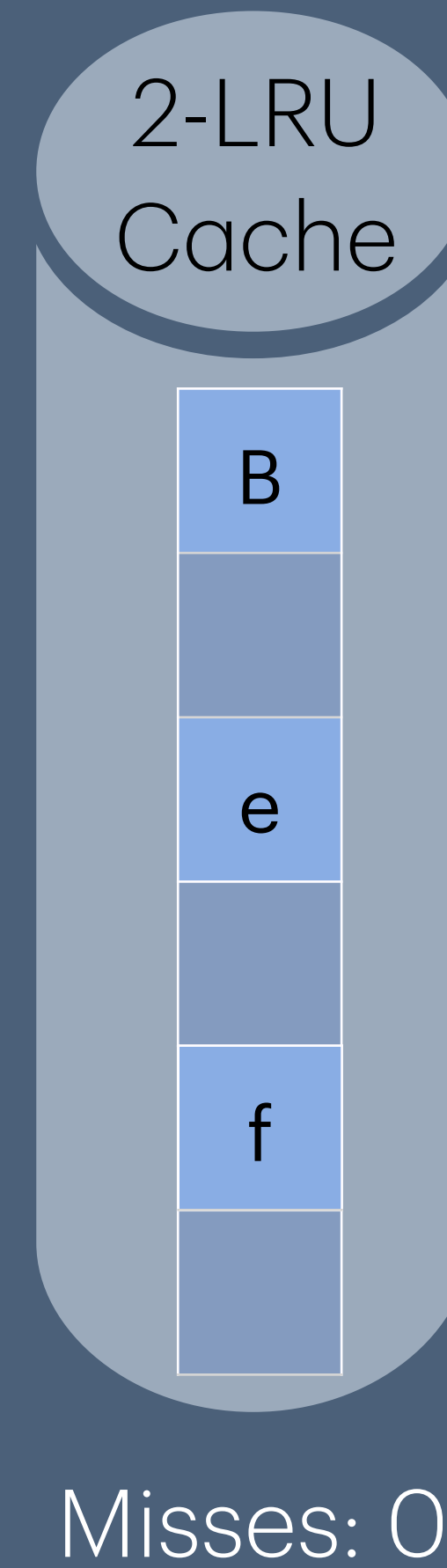
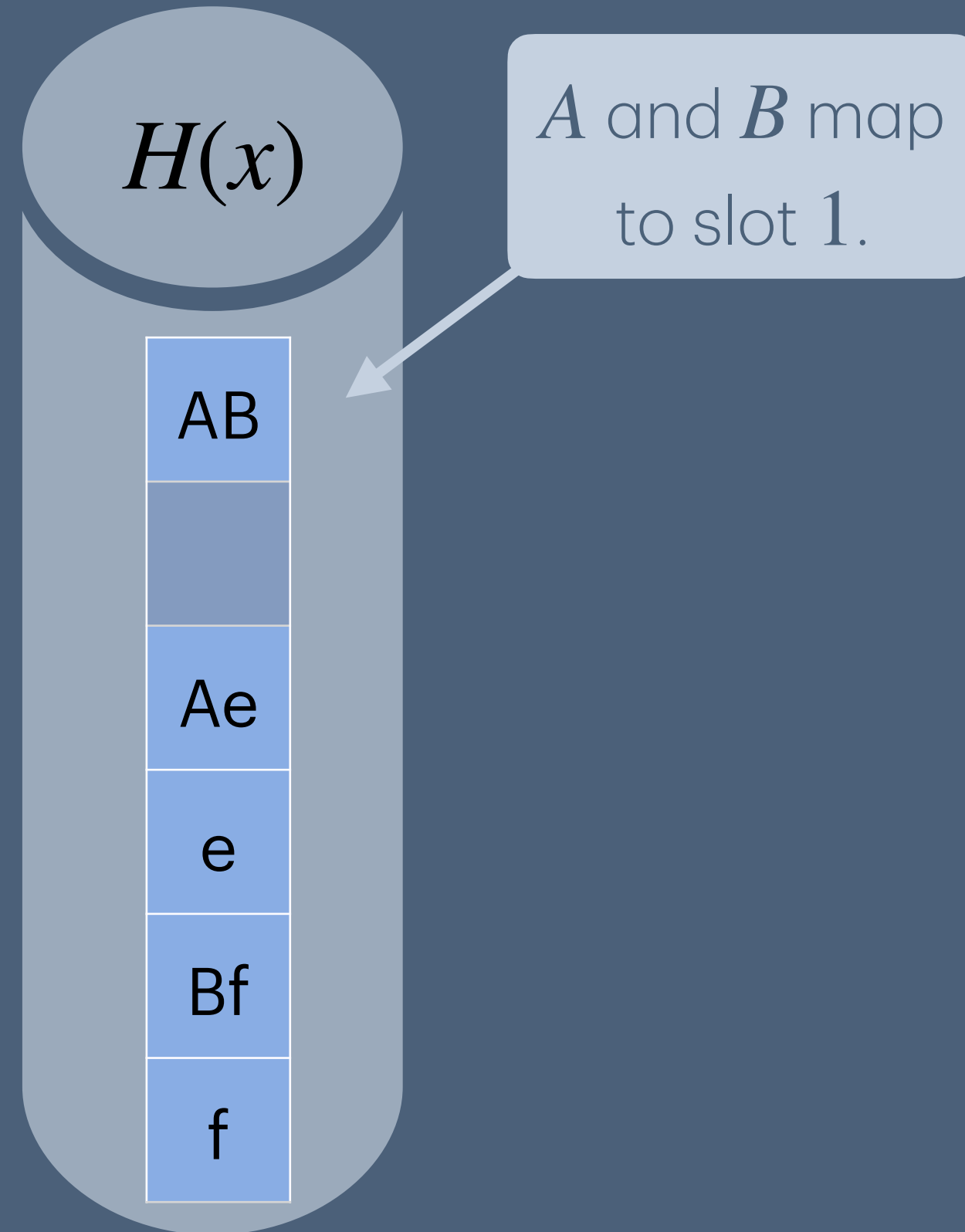
$$\sigma = e, f, A, e, f, B.$$



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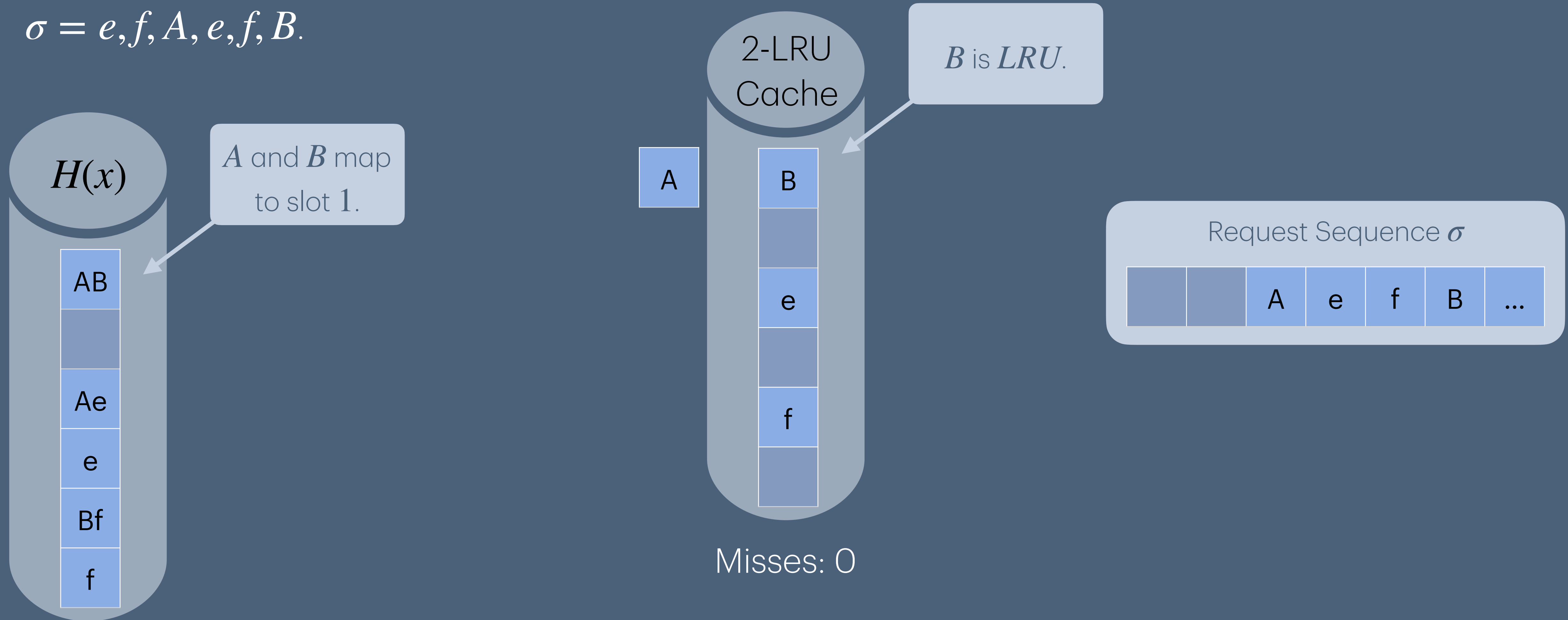
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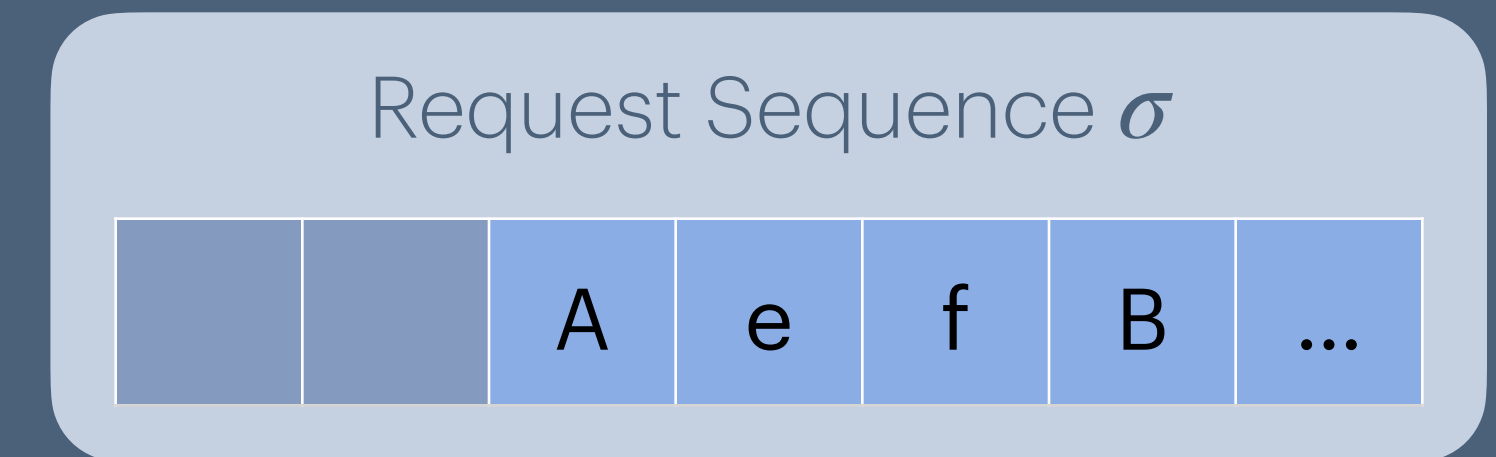
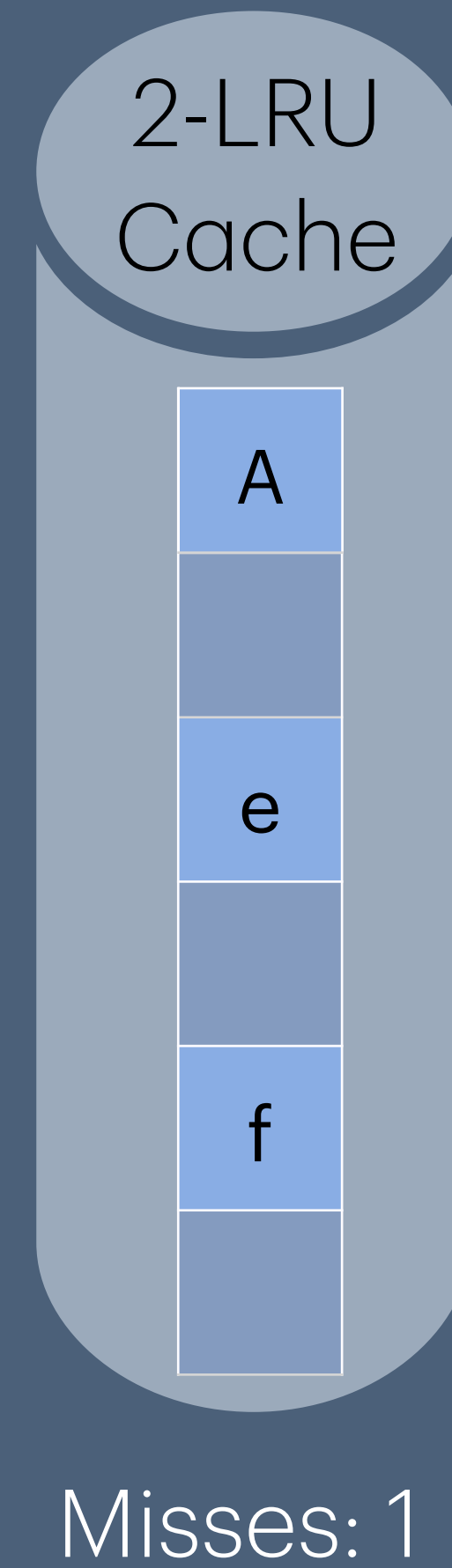
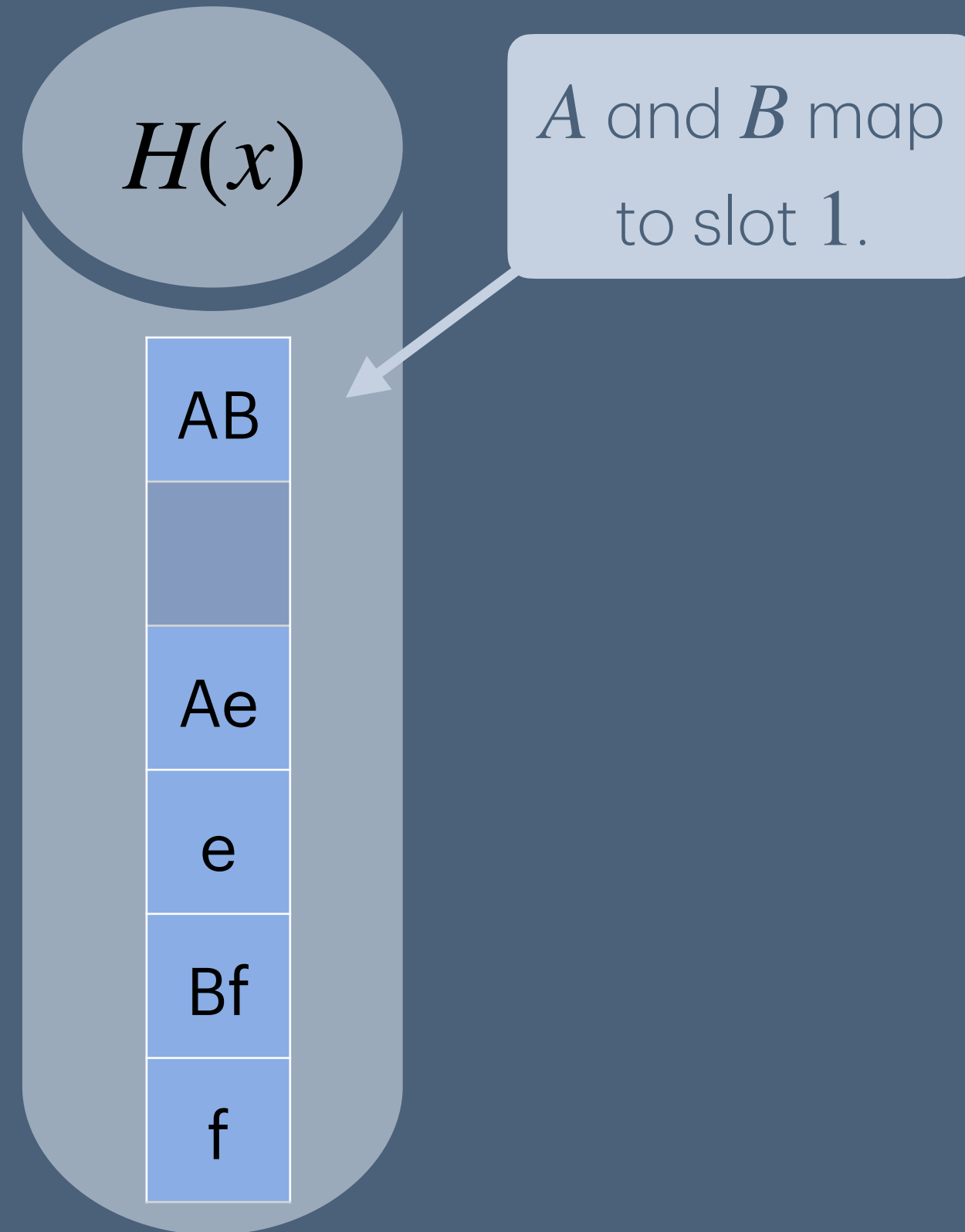
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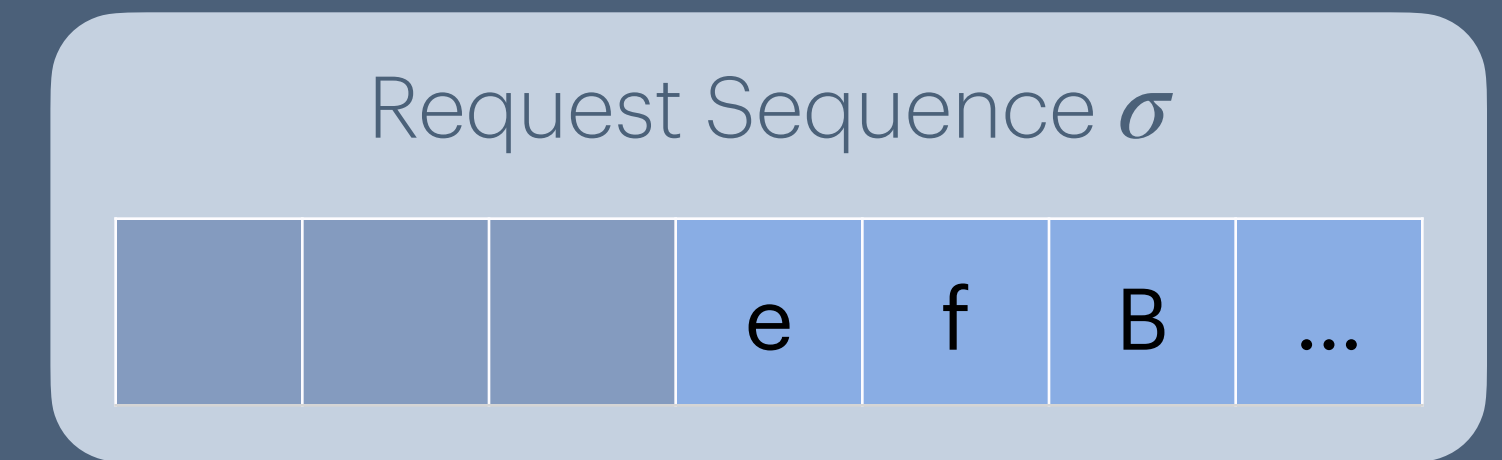
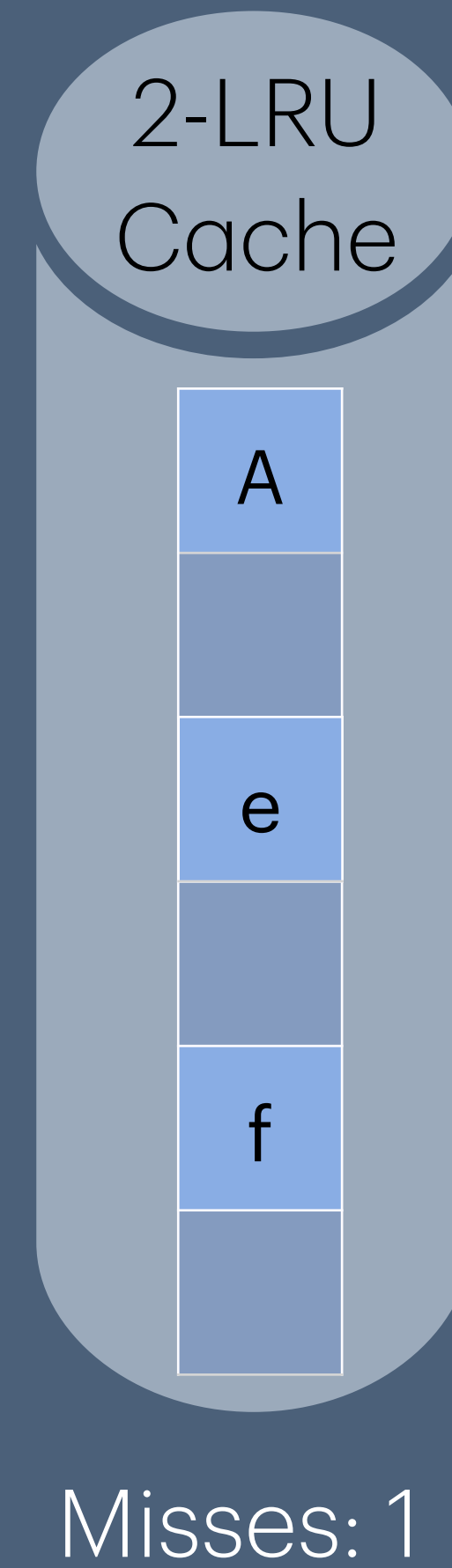
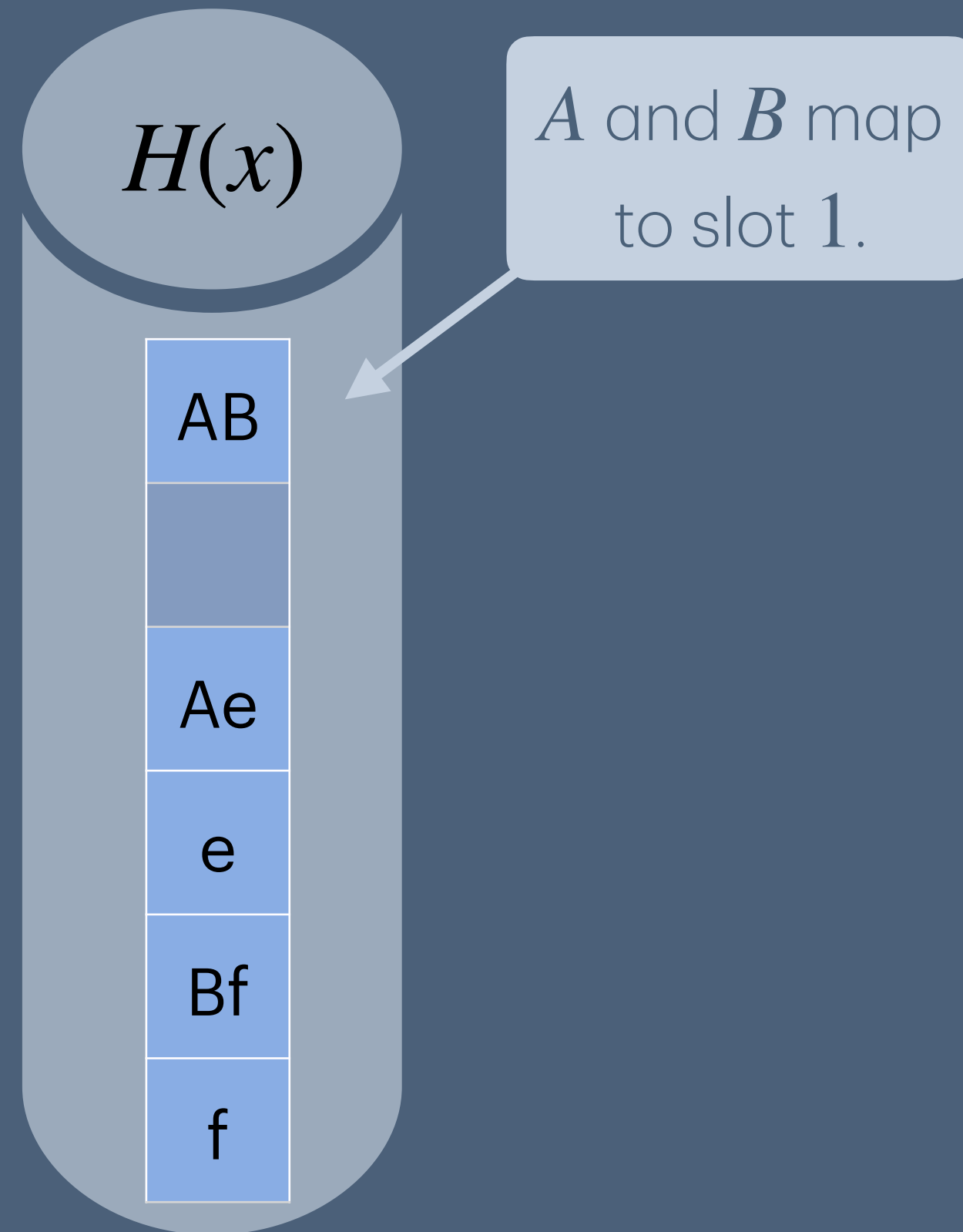
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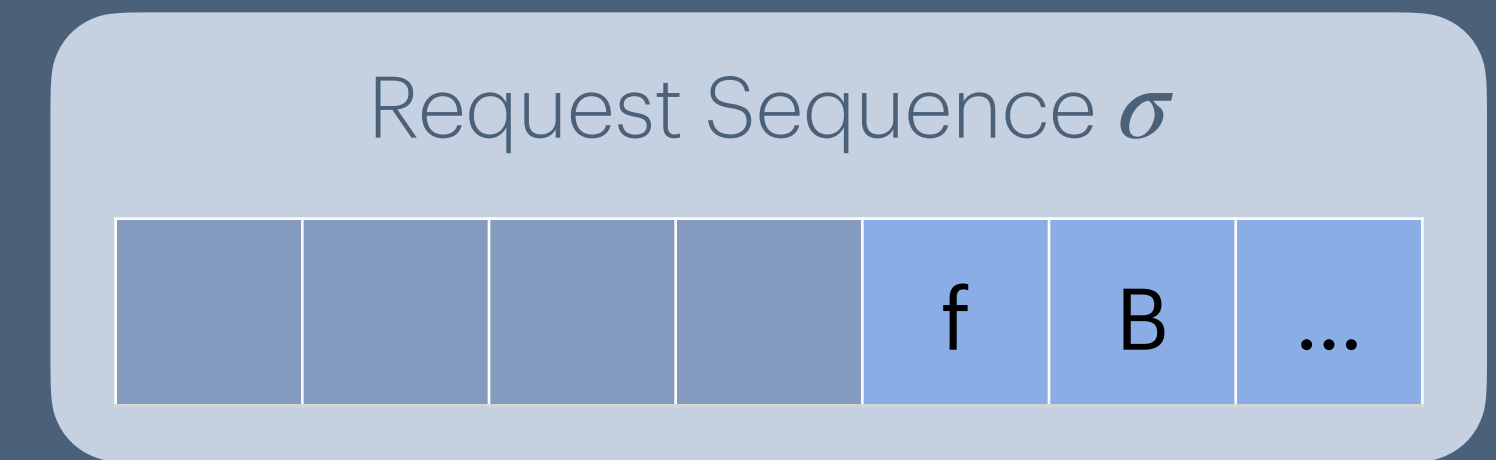
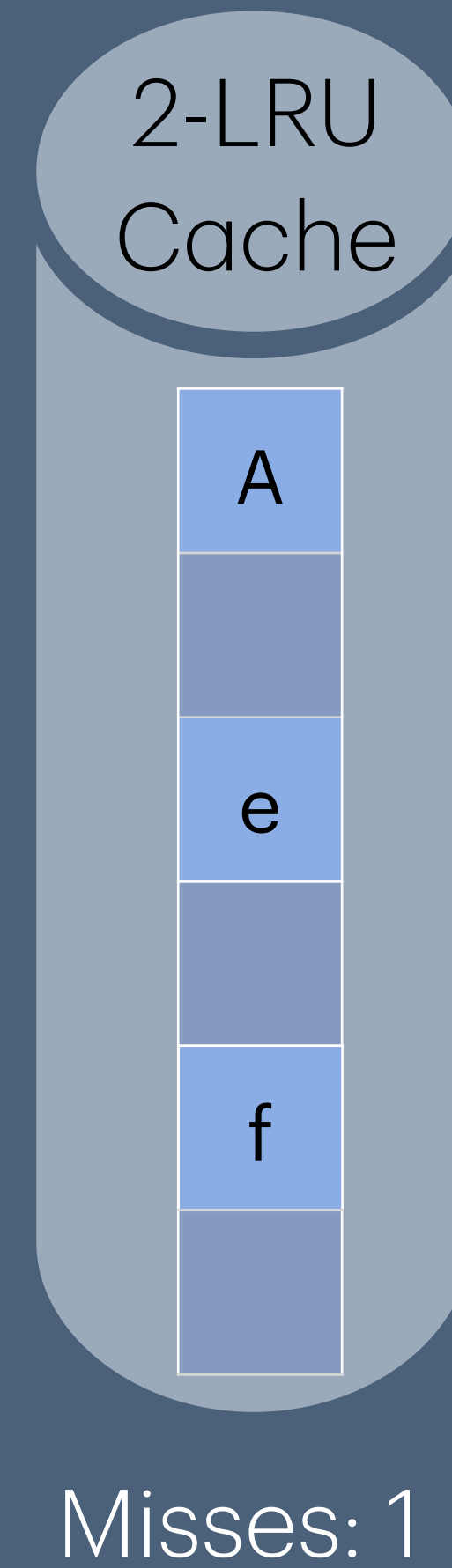
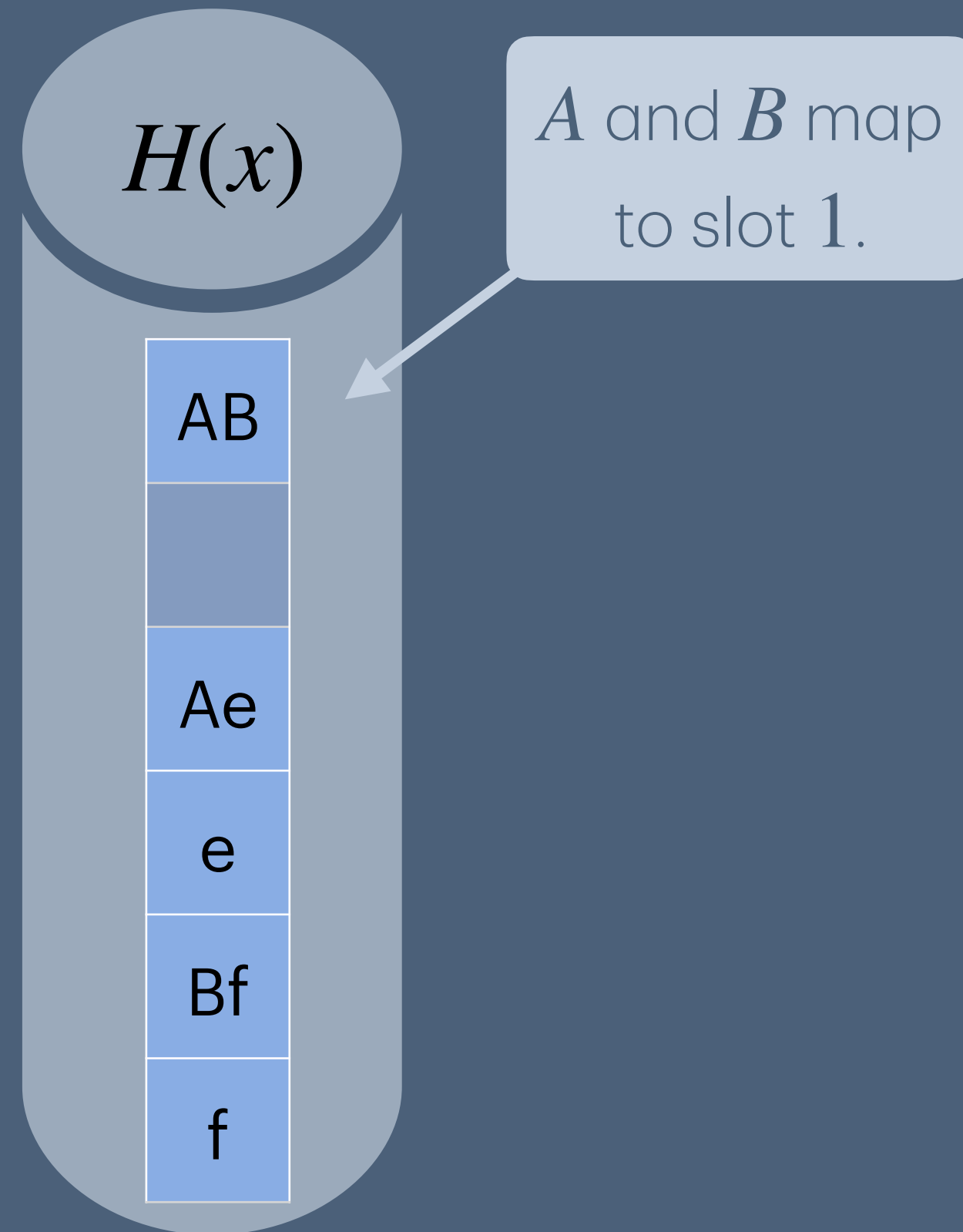
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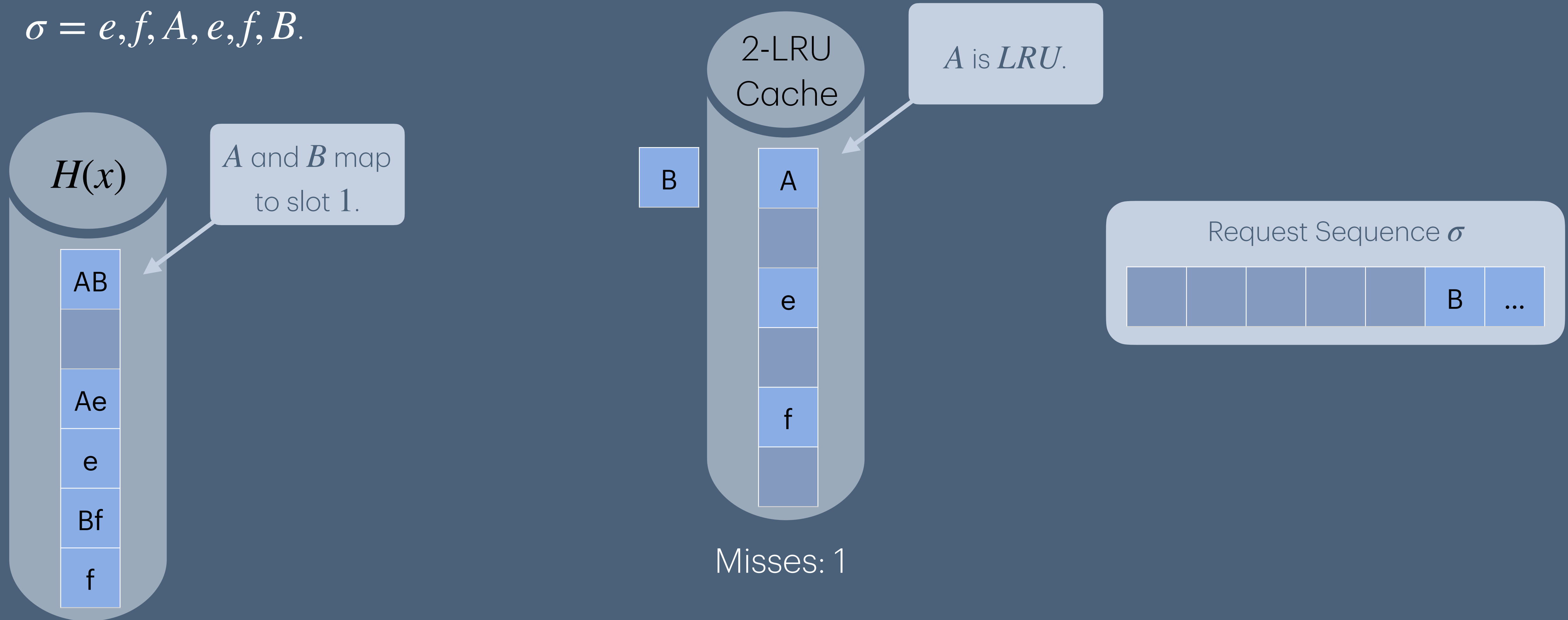
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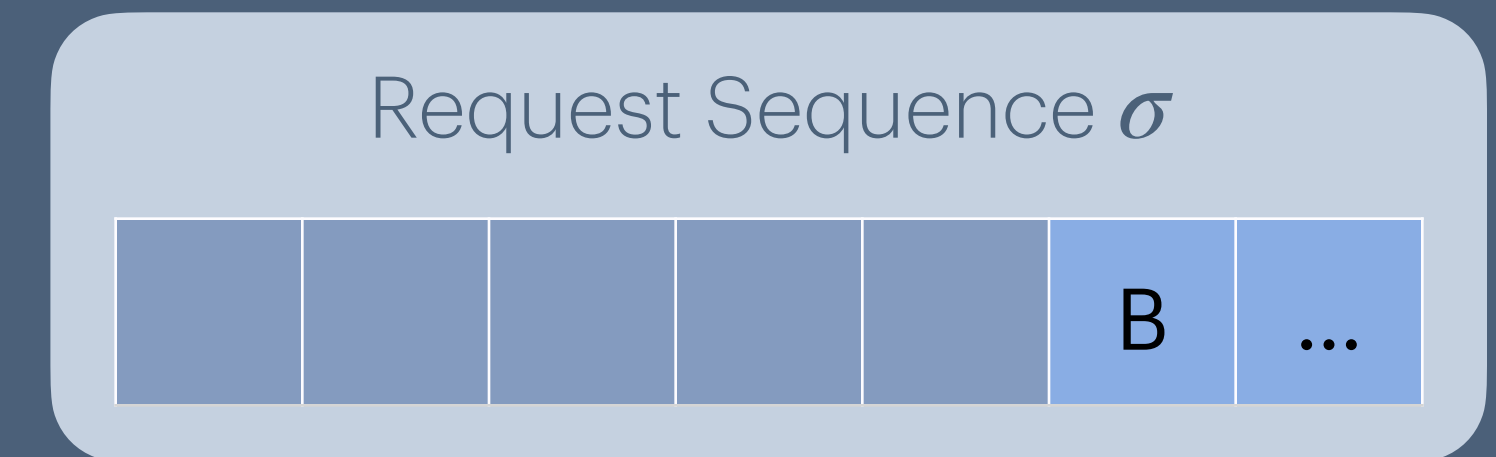
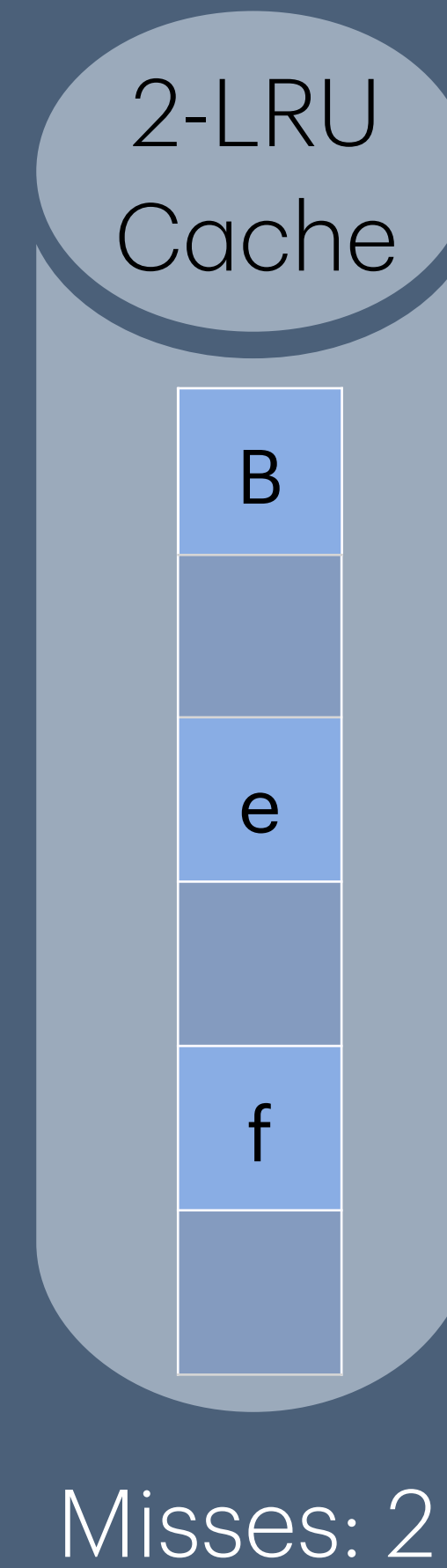
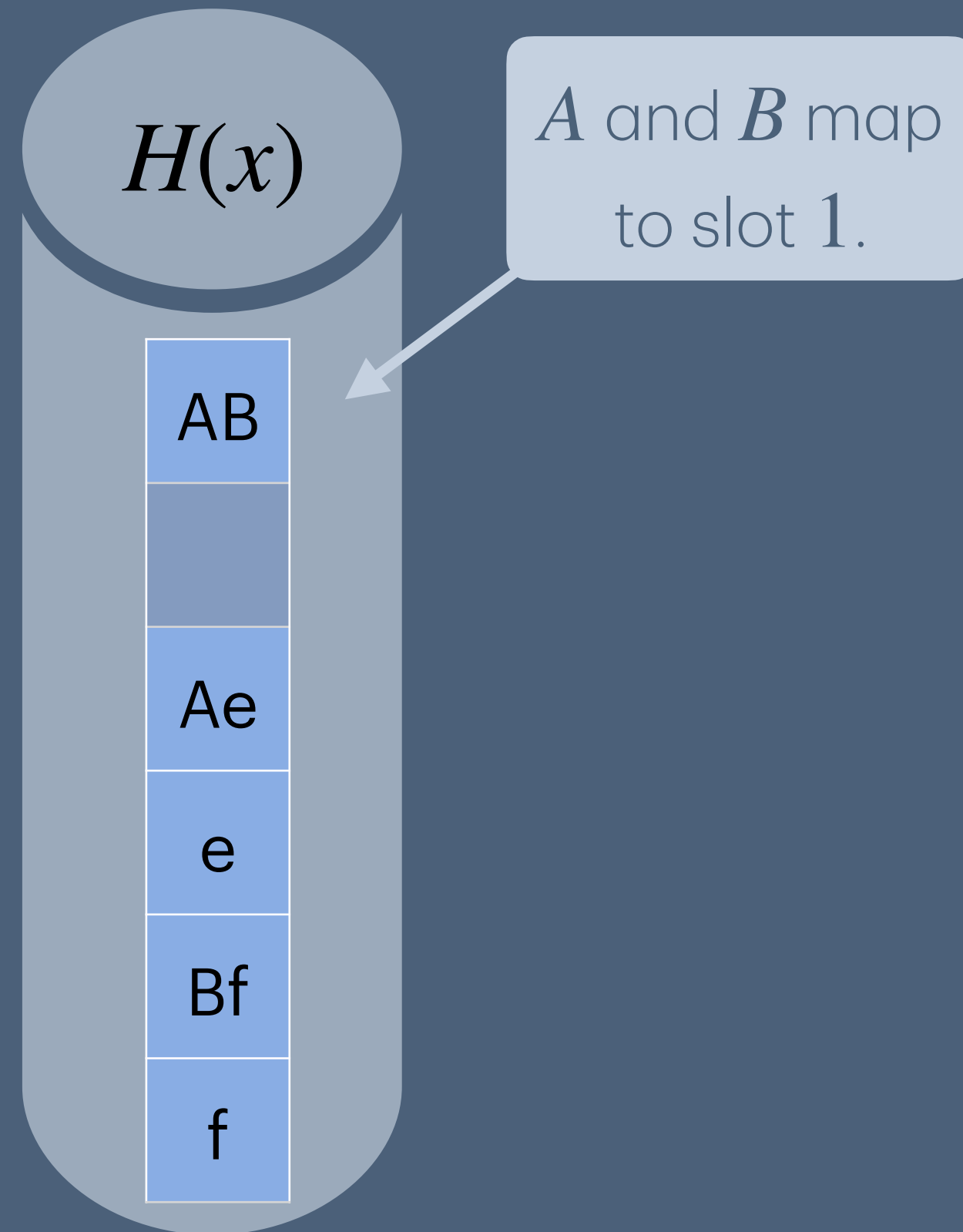
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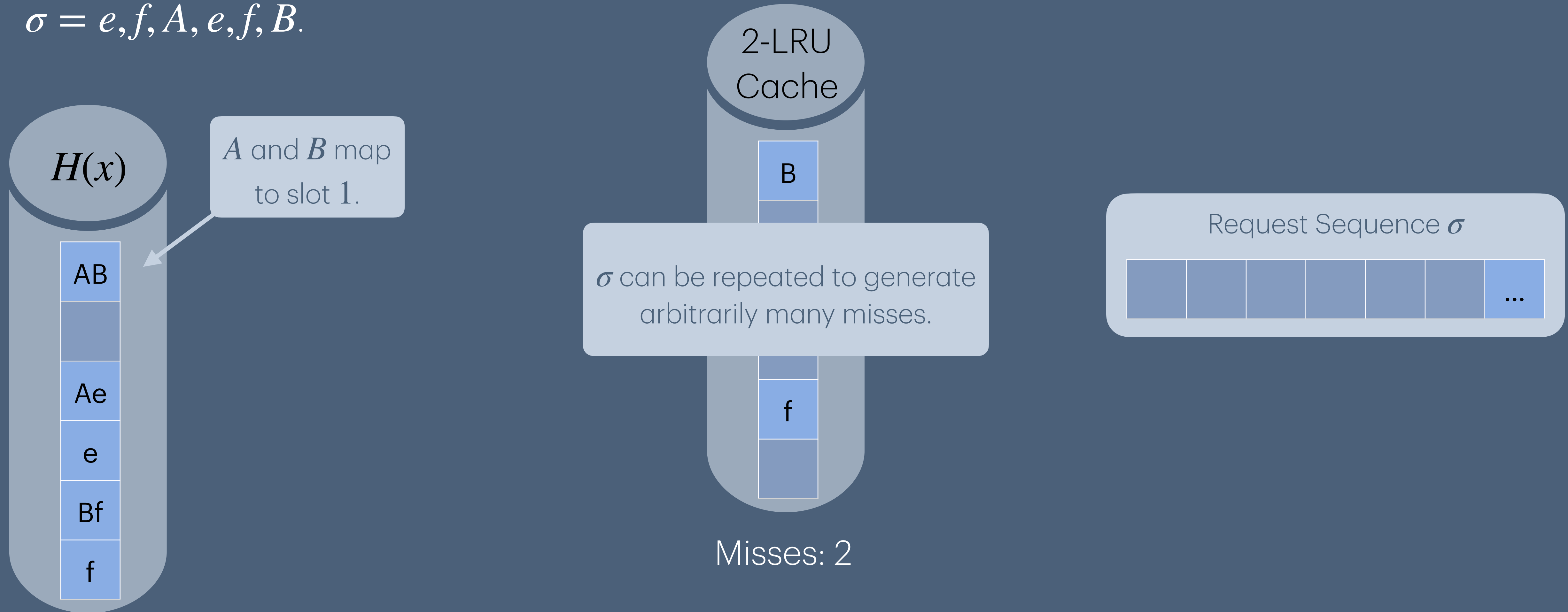
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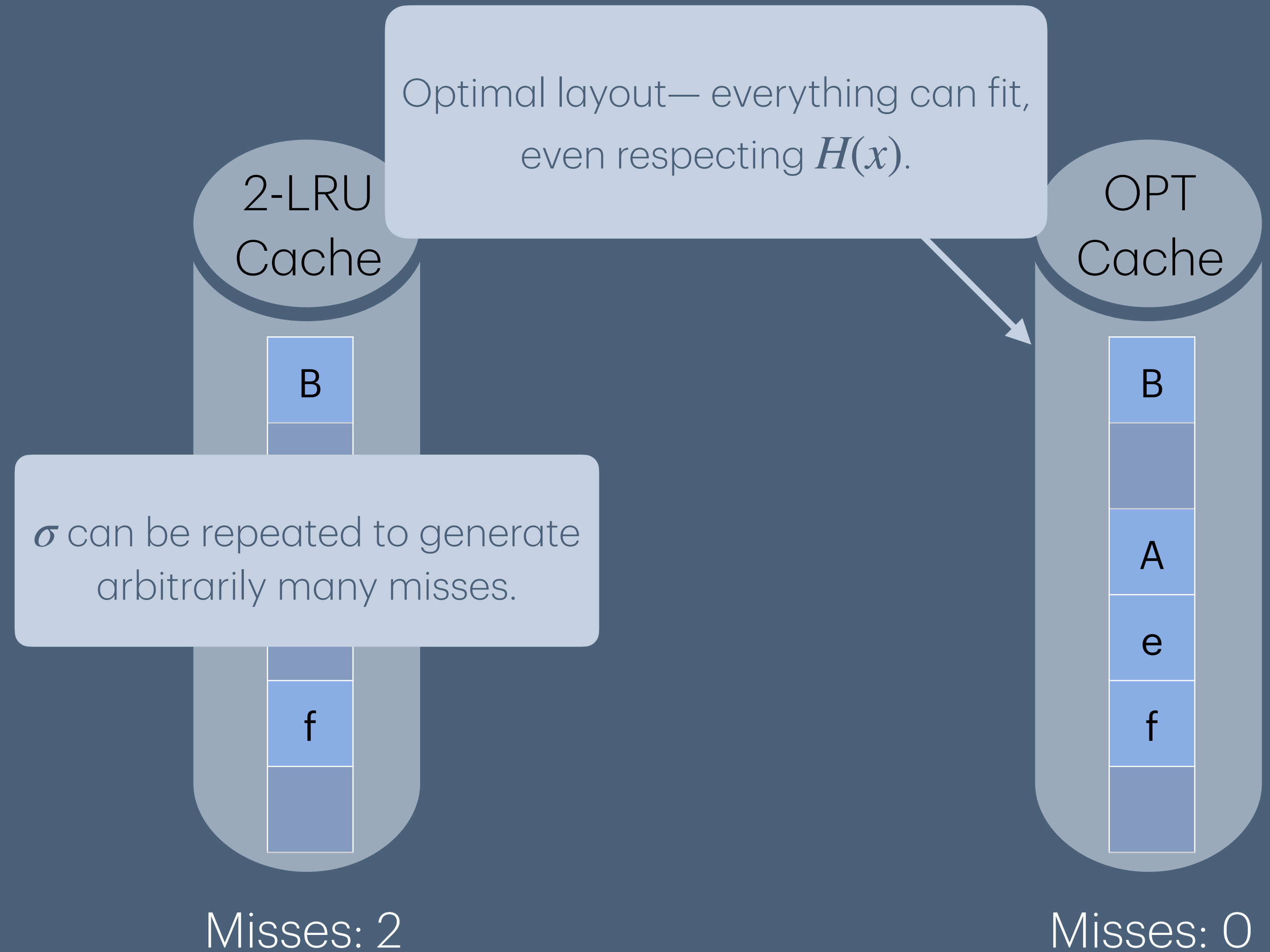
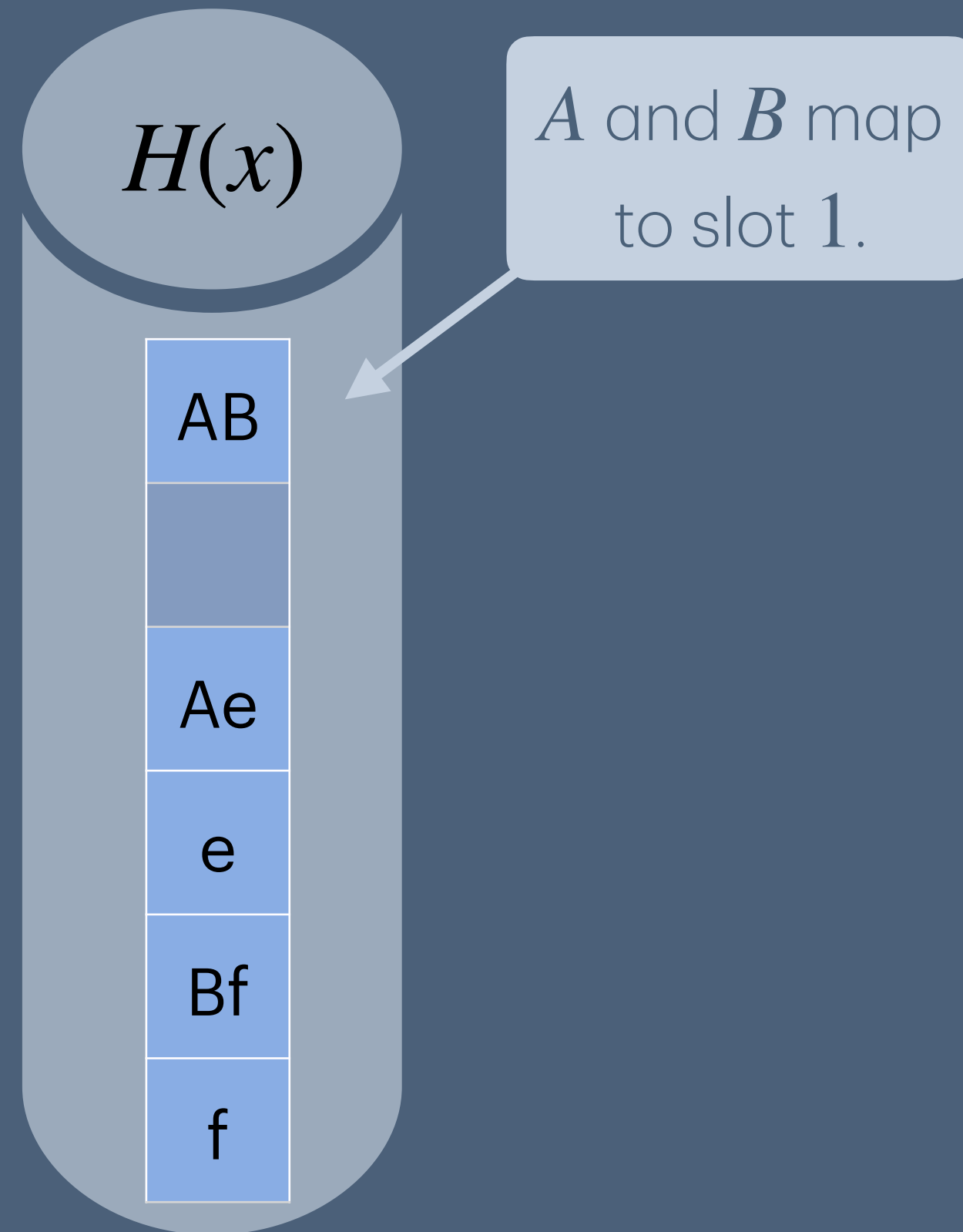
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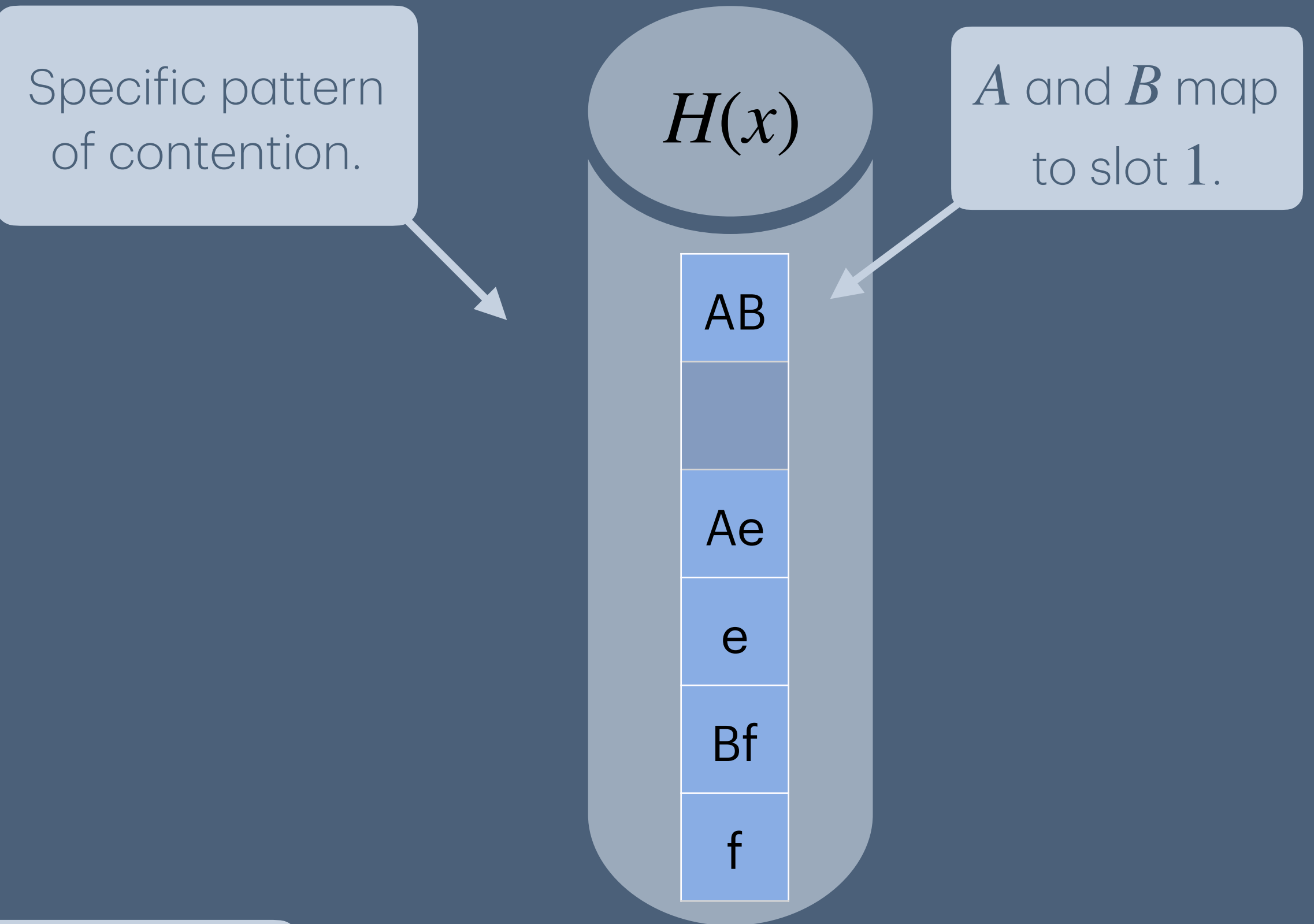


LRU's Downfall: Random Guesses

How can an adversary write down σ when oblivious to $H(x)$?

Guessing— the adversary writes down many candidates for A , B , e and f .

Enough of the candidates will be correct— combine them all.



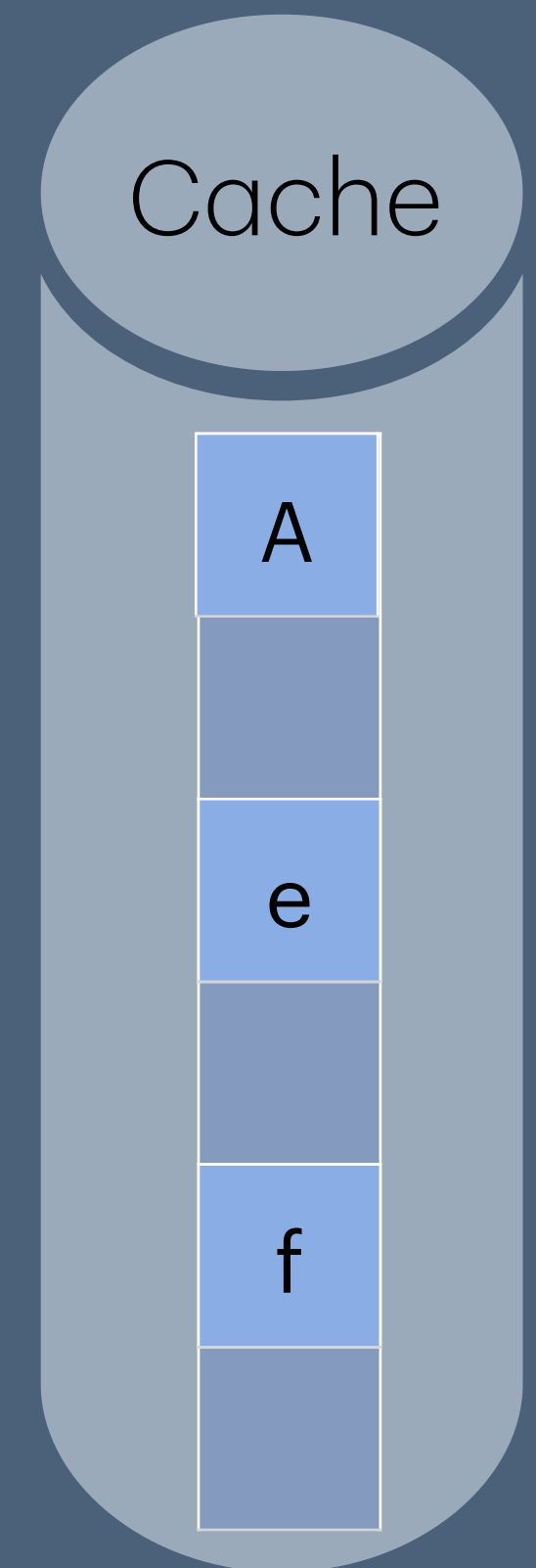
See Paper: Generating candidates for A , B , e and f .

Question: Why is *d-LRU* not competitive while *2-RAND* is?

We'll start with a better way to visualize the caches, especially *2-RAND*.

Visualizing Caches as Graph Orientations

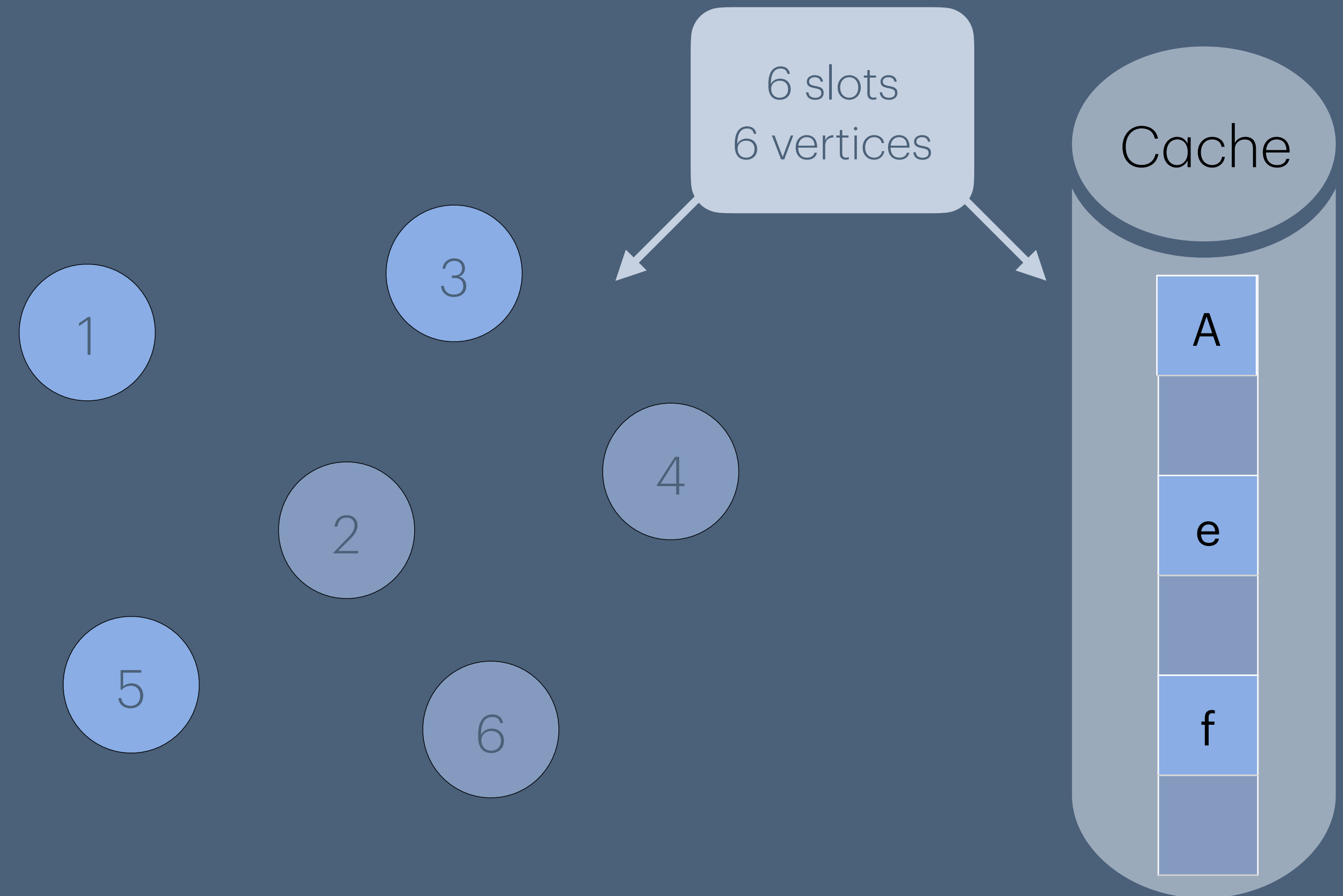
Cache slots become
vertices.



Visualizing Caches as Graph Orientations

Cache slots become
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Pages become **edges**
between $h_1(x)$ and $h_2(x)$.

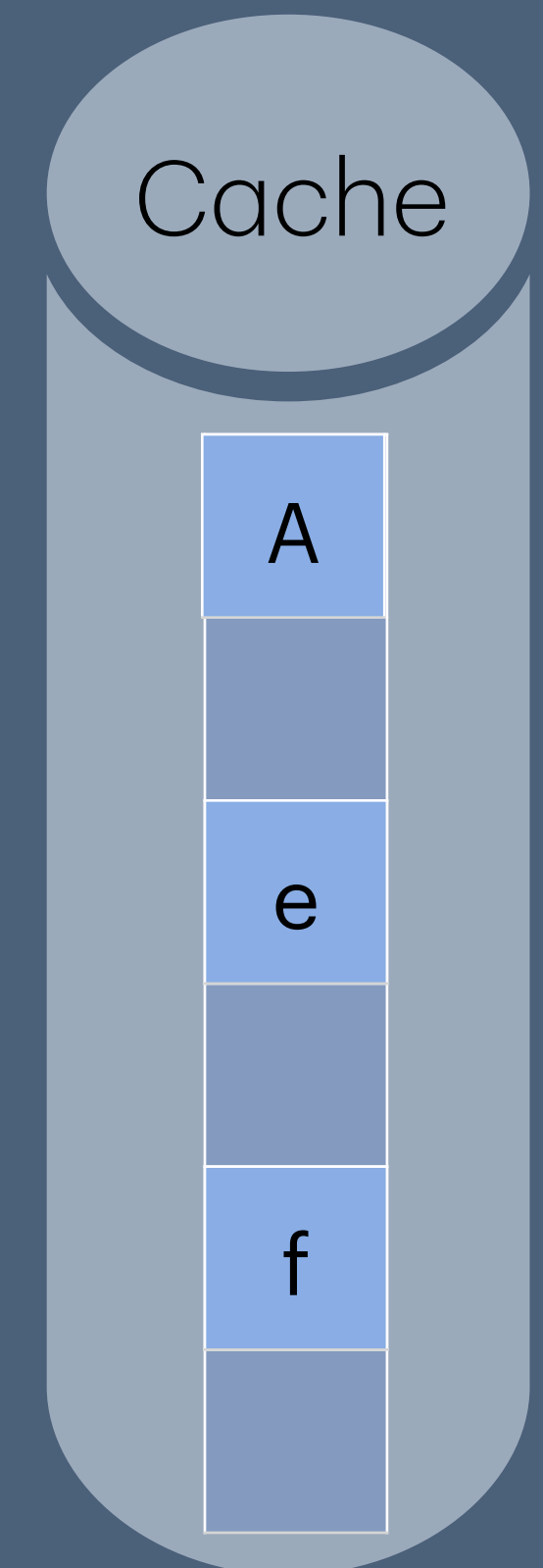
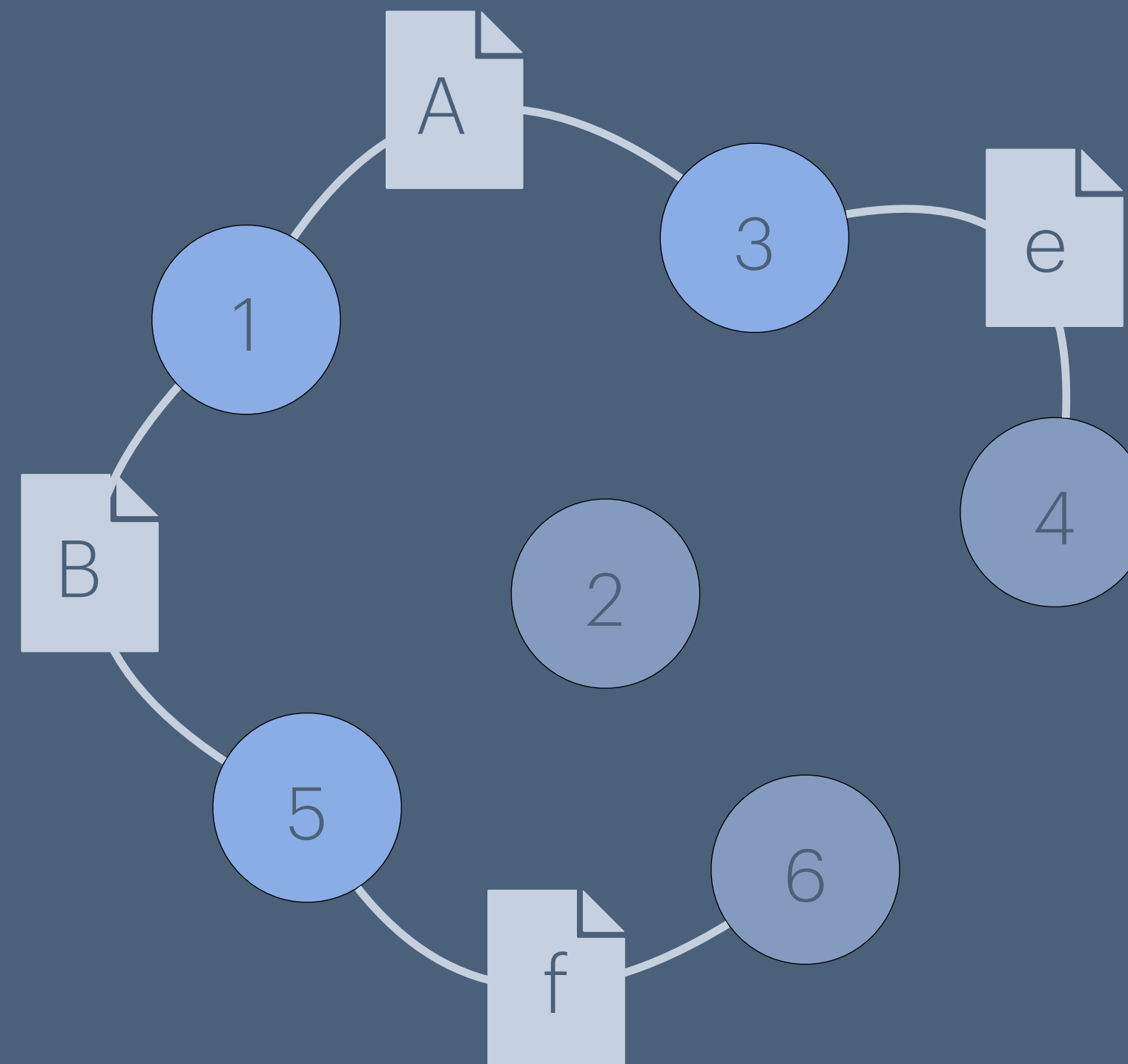


Visualizing Caches as Graph Orientations

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Pages become **edges**
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Edges point where the page **is**
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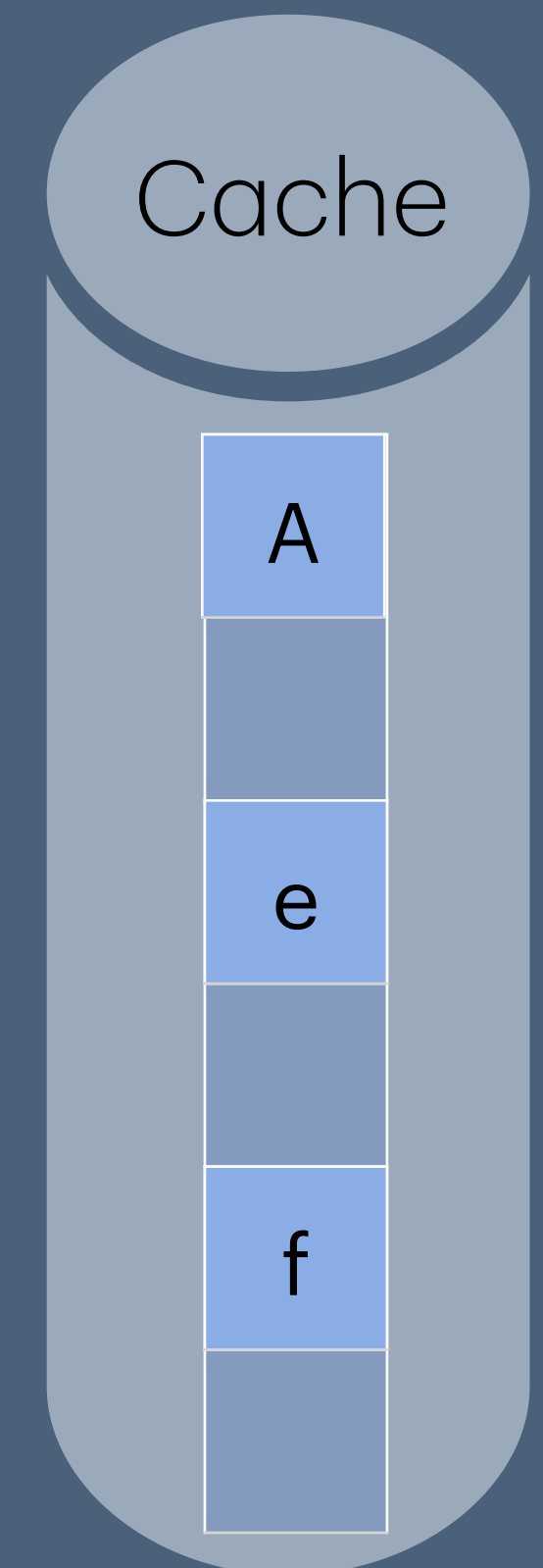
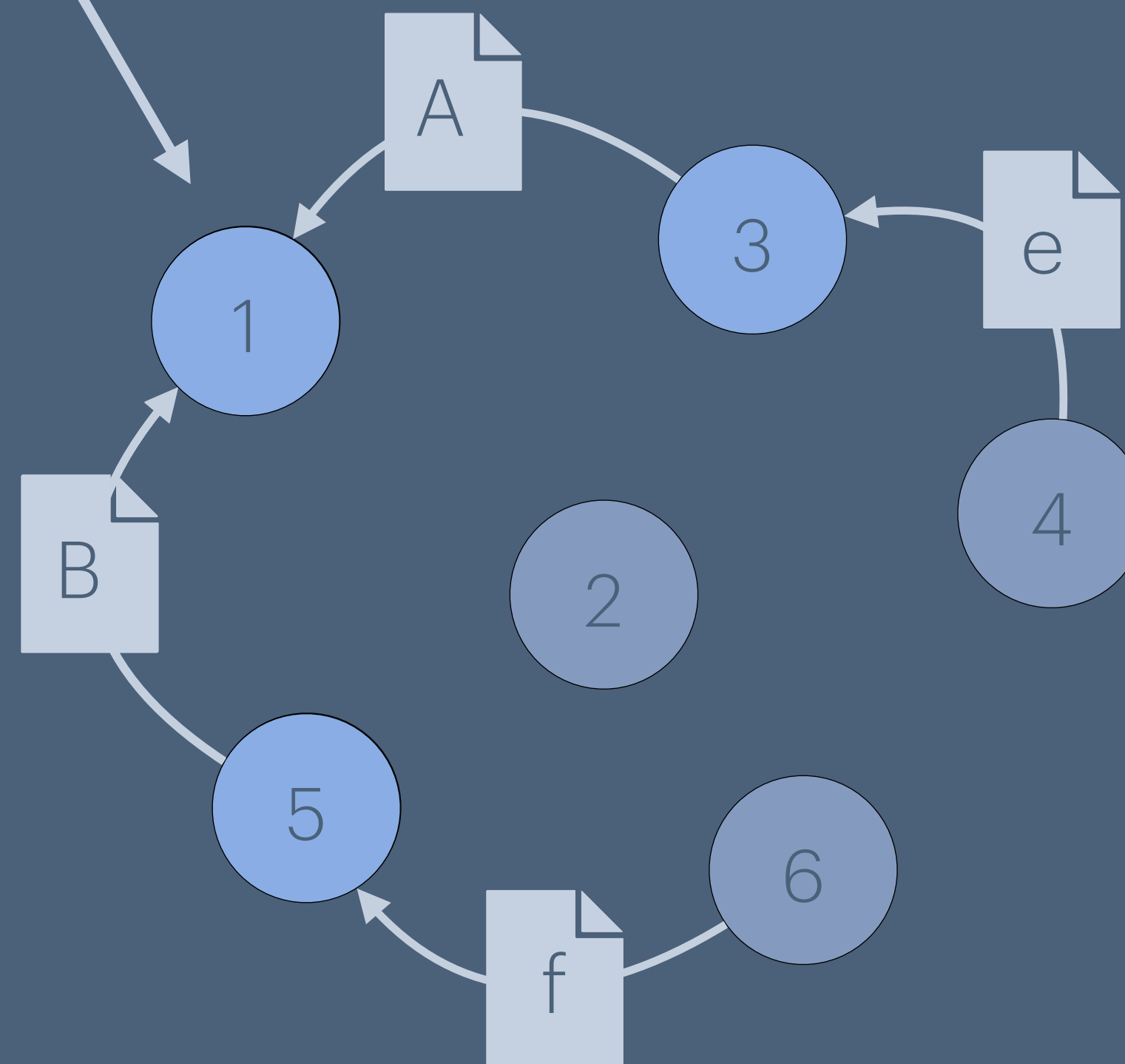
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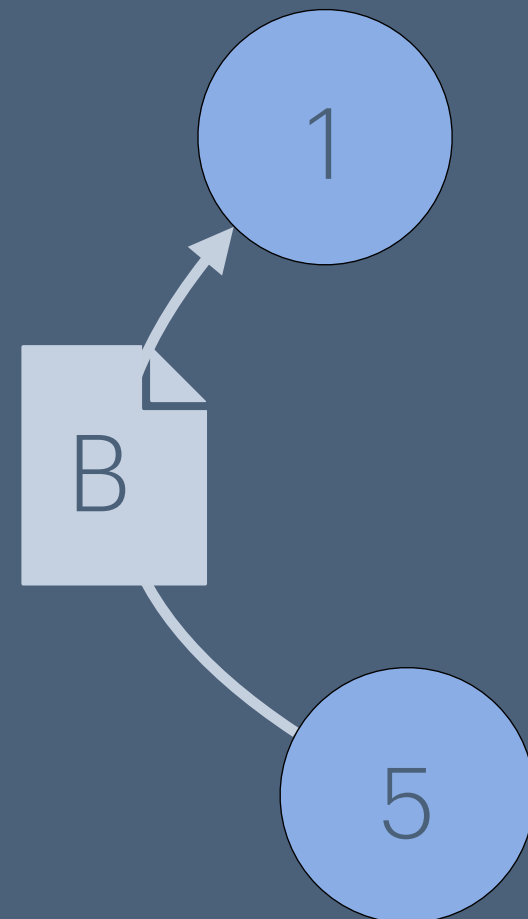
Edges point where the page **is** or **most recently was**.

In-degree 2 or more: some page is *not* in cache.

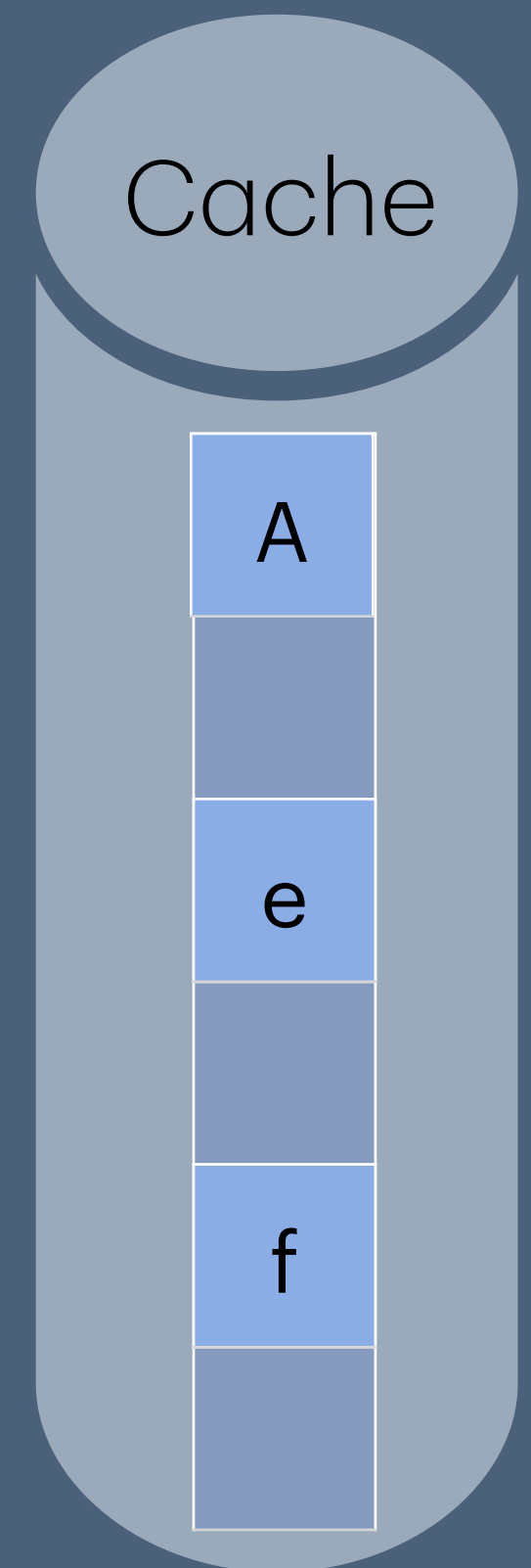


Visualizing Caches as Graph Orientations

Consider a miss on *B*.

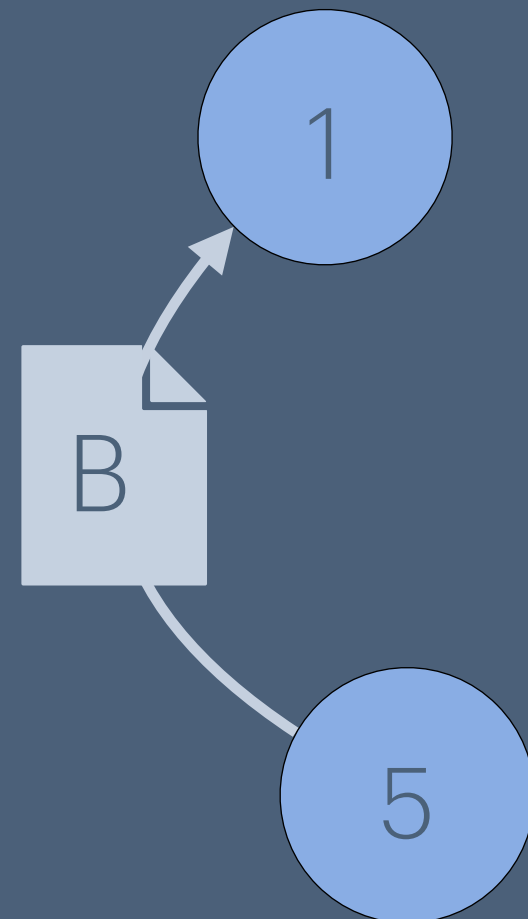


Not in cache.



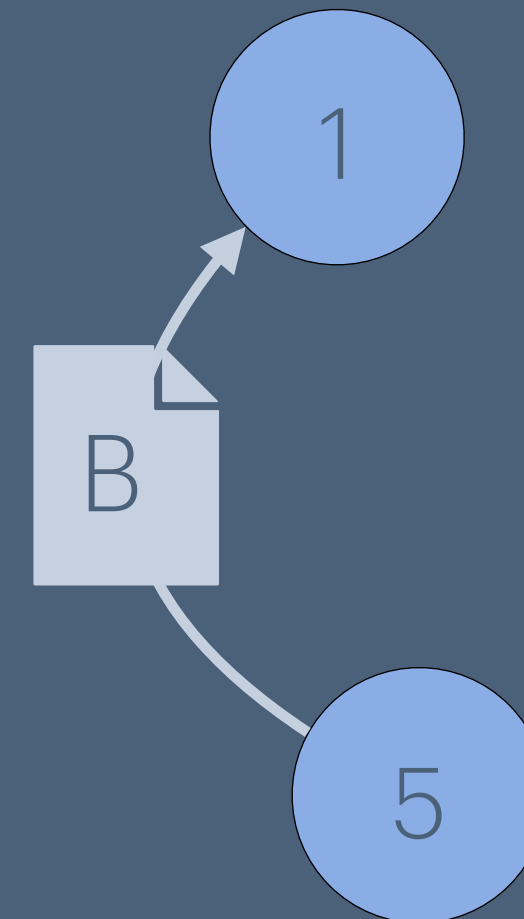
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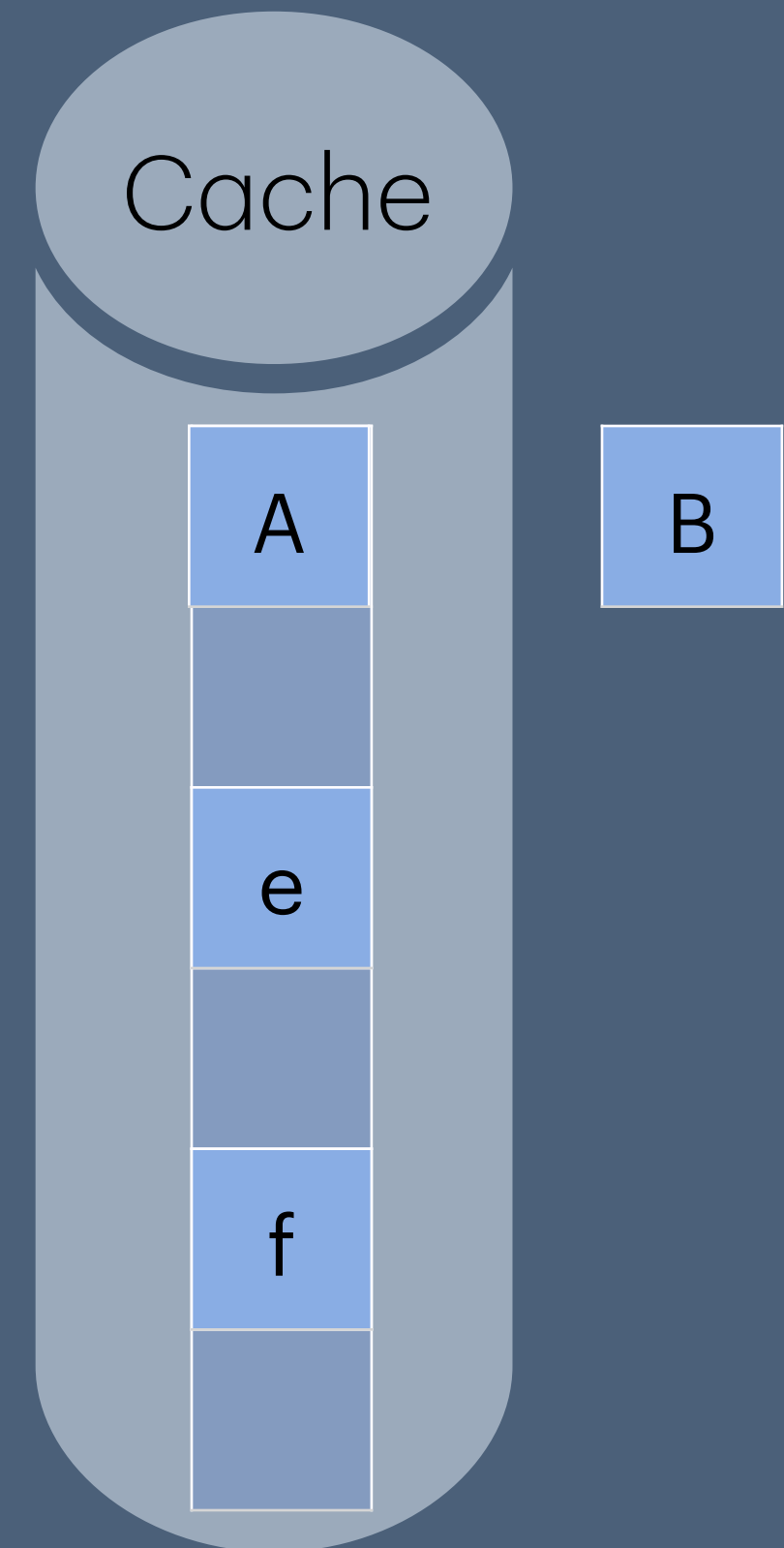


Not in cache.

Slot 1.

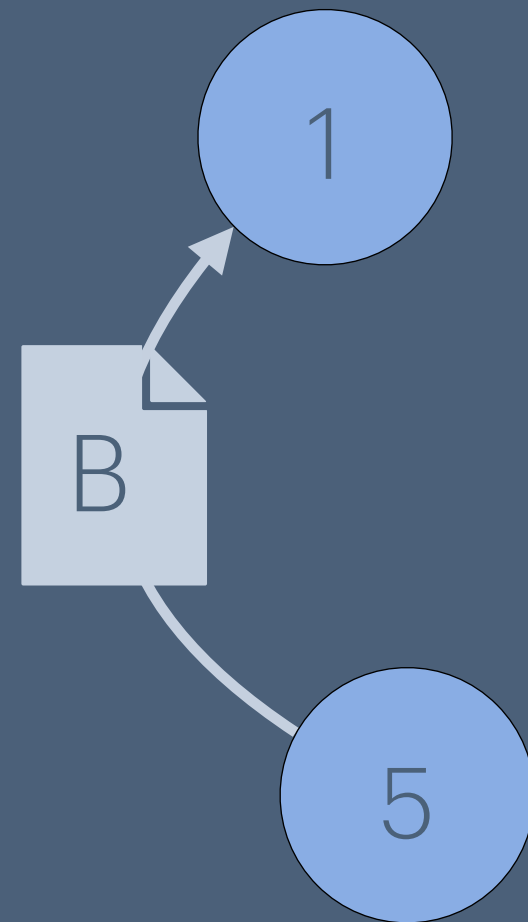


Cache



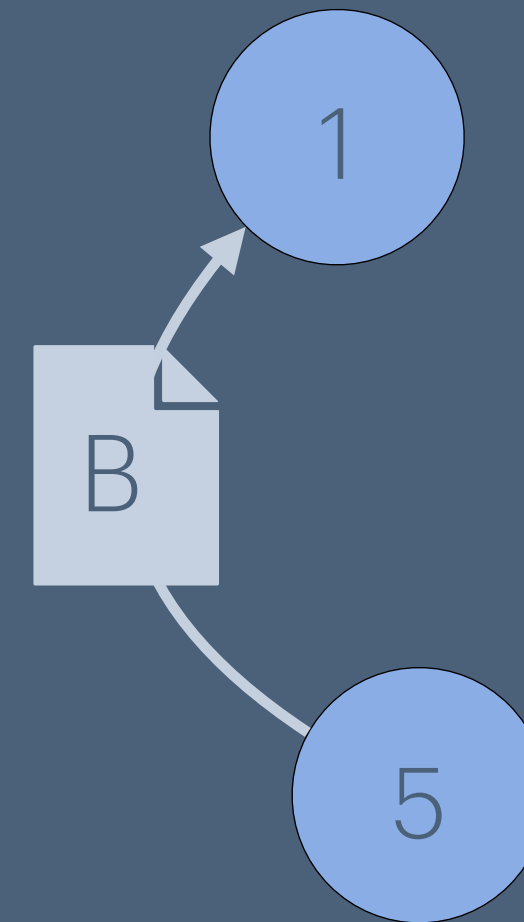
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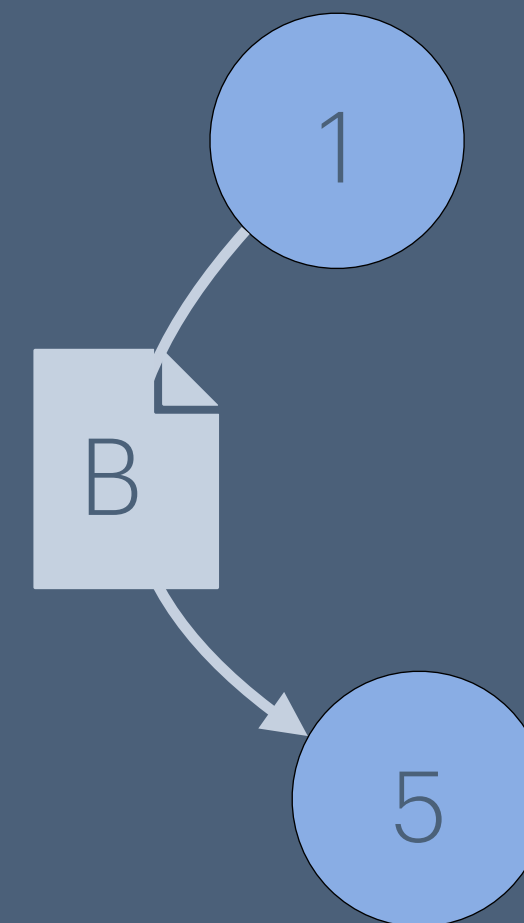


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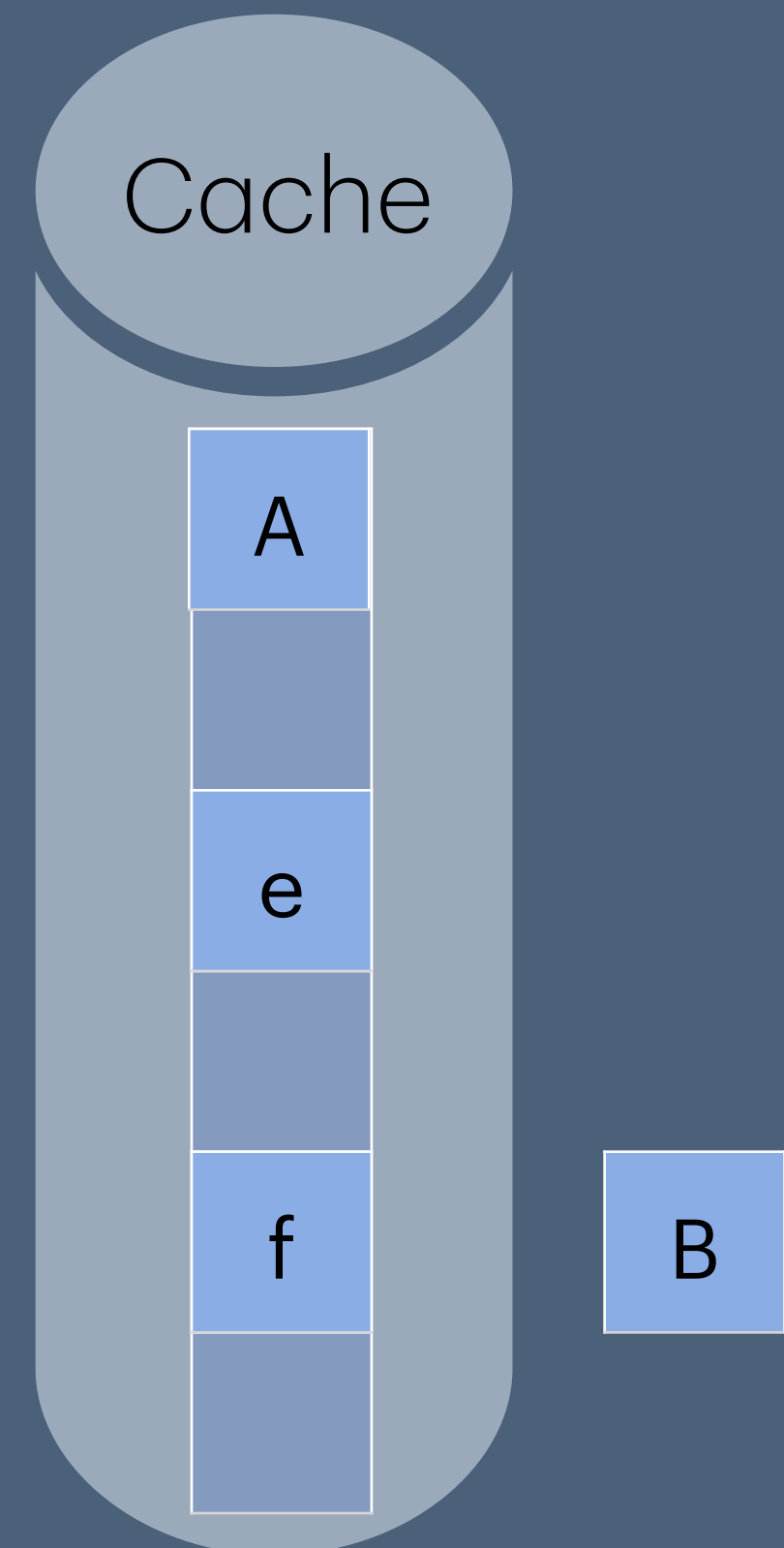
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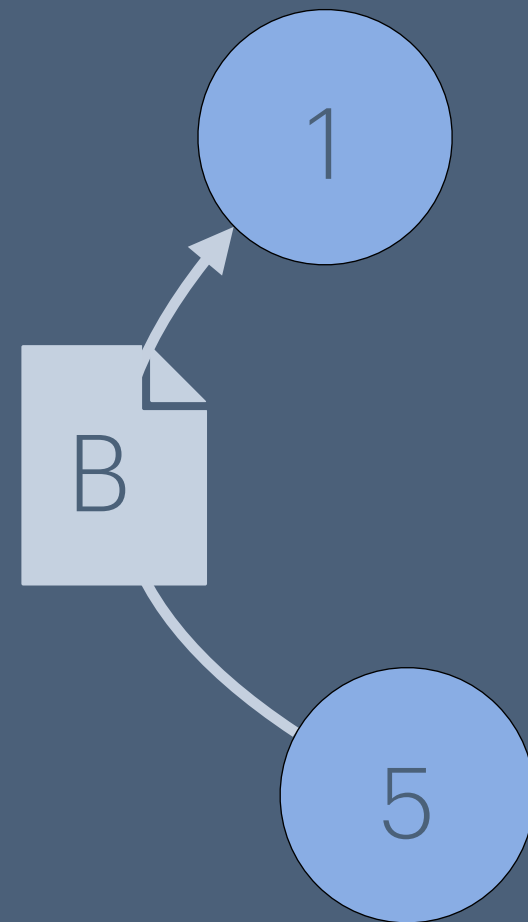
Slot 5.



Cache

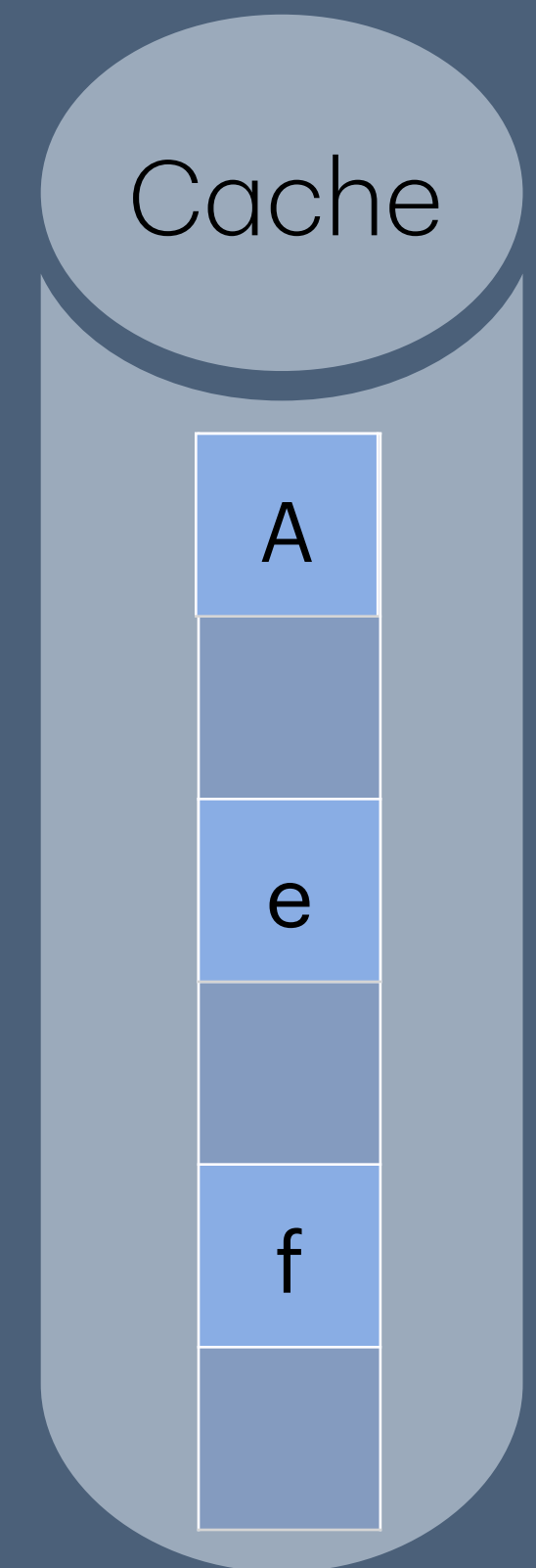


Visualizing Caches as Graph Orientations



If every vertex has in-degree 1, then **every considered page is in cache**— every request for these pages will be a hit.

If every **connected component** is oriented as above, then the graph is oriented as above.



2-RAND: Orienting Sparse Graphs

$O(1)$ Space Augmentation

$O(1)$ Competitive

OPT cache is size $n/100$.
2-RAND cache is size n .

Goal: Prove *2-RAND* incurs
 $O(n)$ misses per phase.

Define a phase as *OPT* incurring $n/100$ misses.

It follows that there's $n/50$ pages per phase:

$n/100$ pages already in *OPT*.

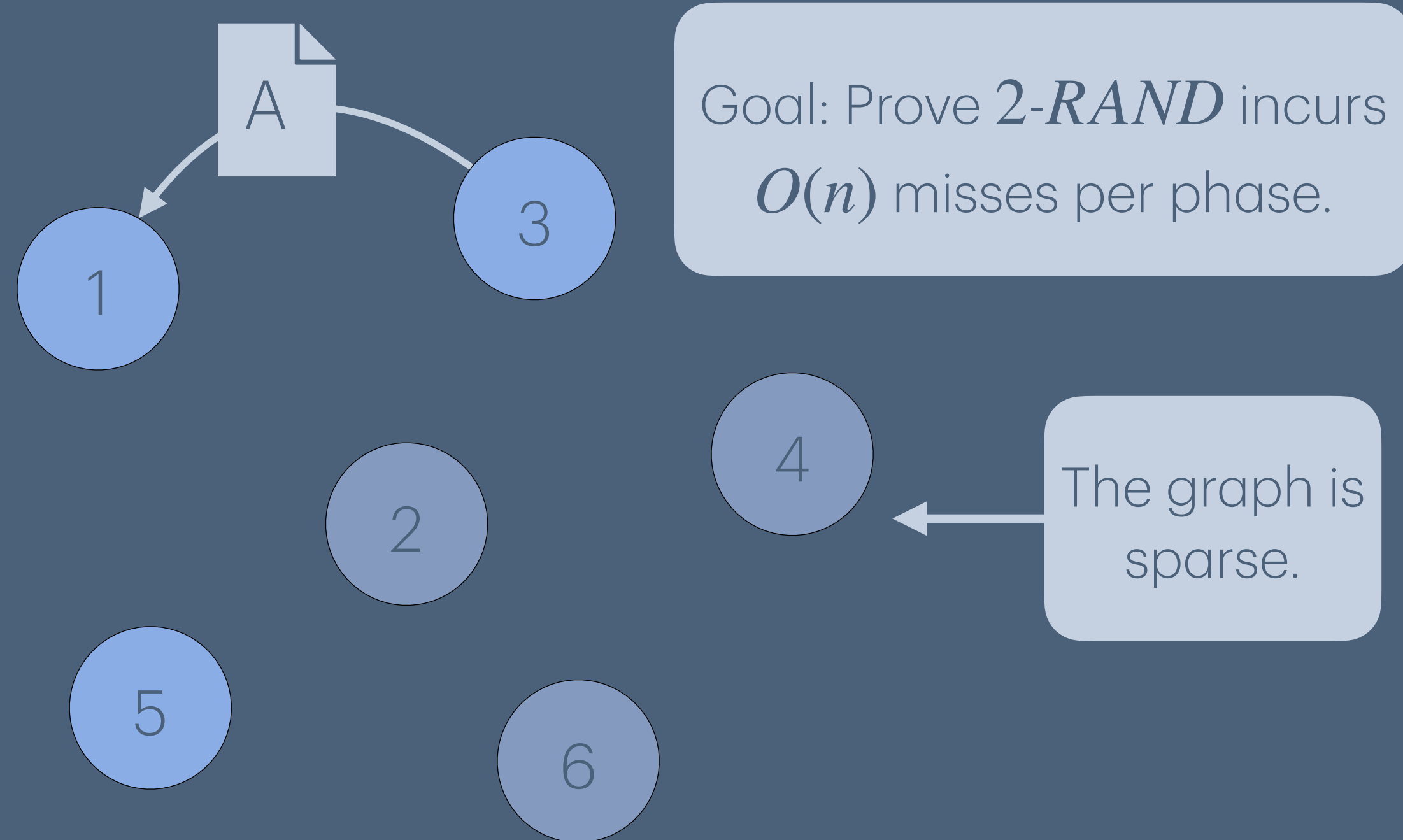
$n/100$ pages from misses.

The graph is
sparse.

2-RAND: Orienting Sparse Graphs

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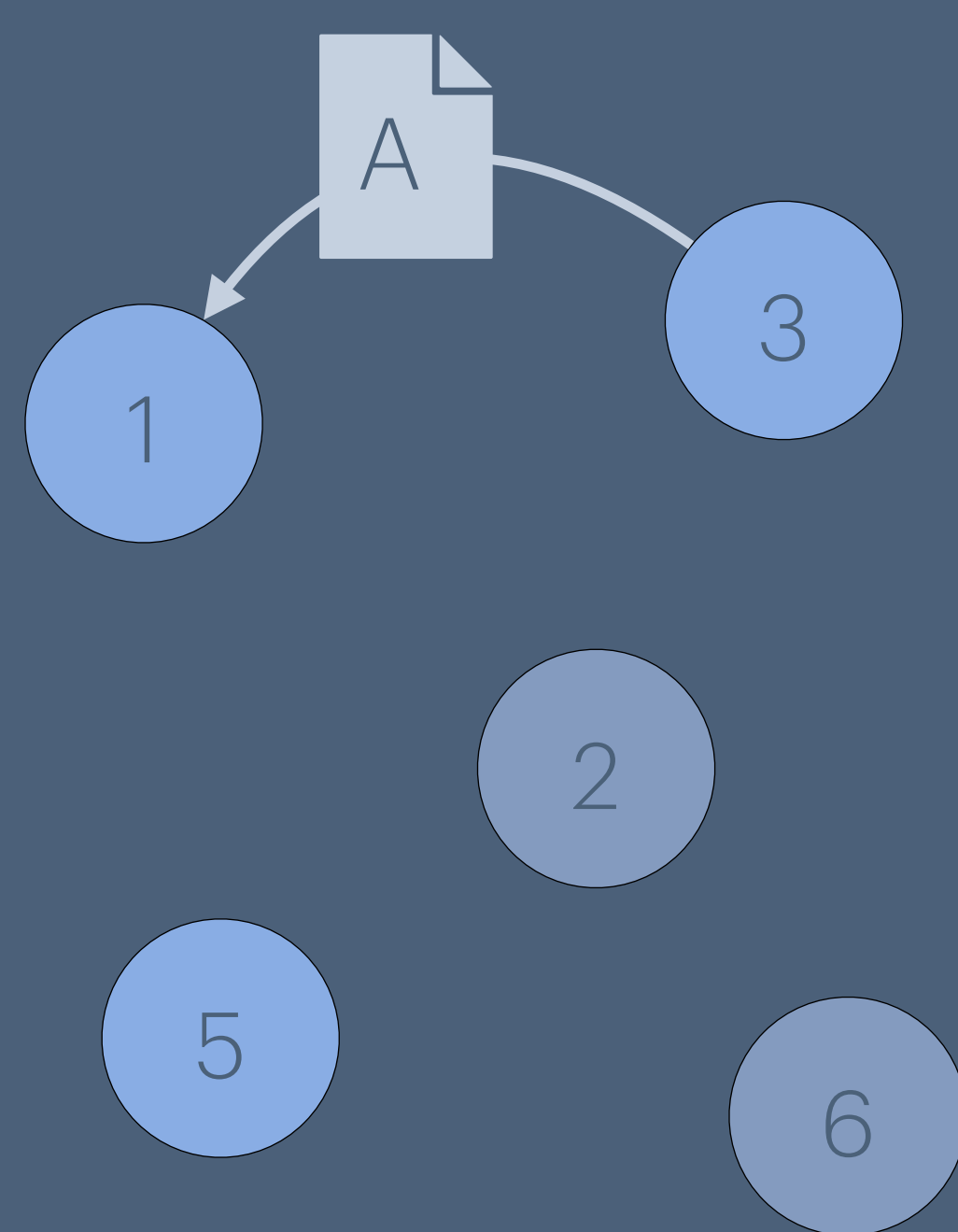
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2-RAND: Orienting Sparse Graphs

$O(1)$ Space Augmentation

$O(1)$ Competitive



Goal: Prove *2-RAND* incurs $O(n)$ misses per phase.

WTS: Each component is oriented within $O(1)$ evictions.

$$E[w(C)] := \sum_{i=1}^n w(C) \cdot \Pr[|C| = i] \in O(1)$$

$w(C)$: Cost (evictions) for *2-RAND* to randomly walk to a satisfying orientation (out of the $2^{|C|}$ orientations).

Components are small:

$$\Pr[|C| \geq i] \leq \frac{1}{4^{i-2}}.$$

This cost is dominated by that probability.

See Paper for analysis.

Results

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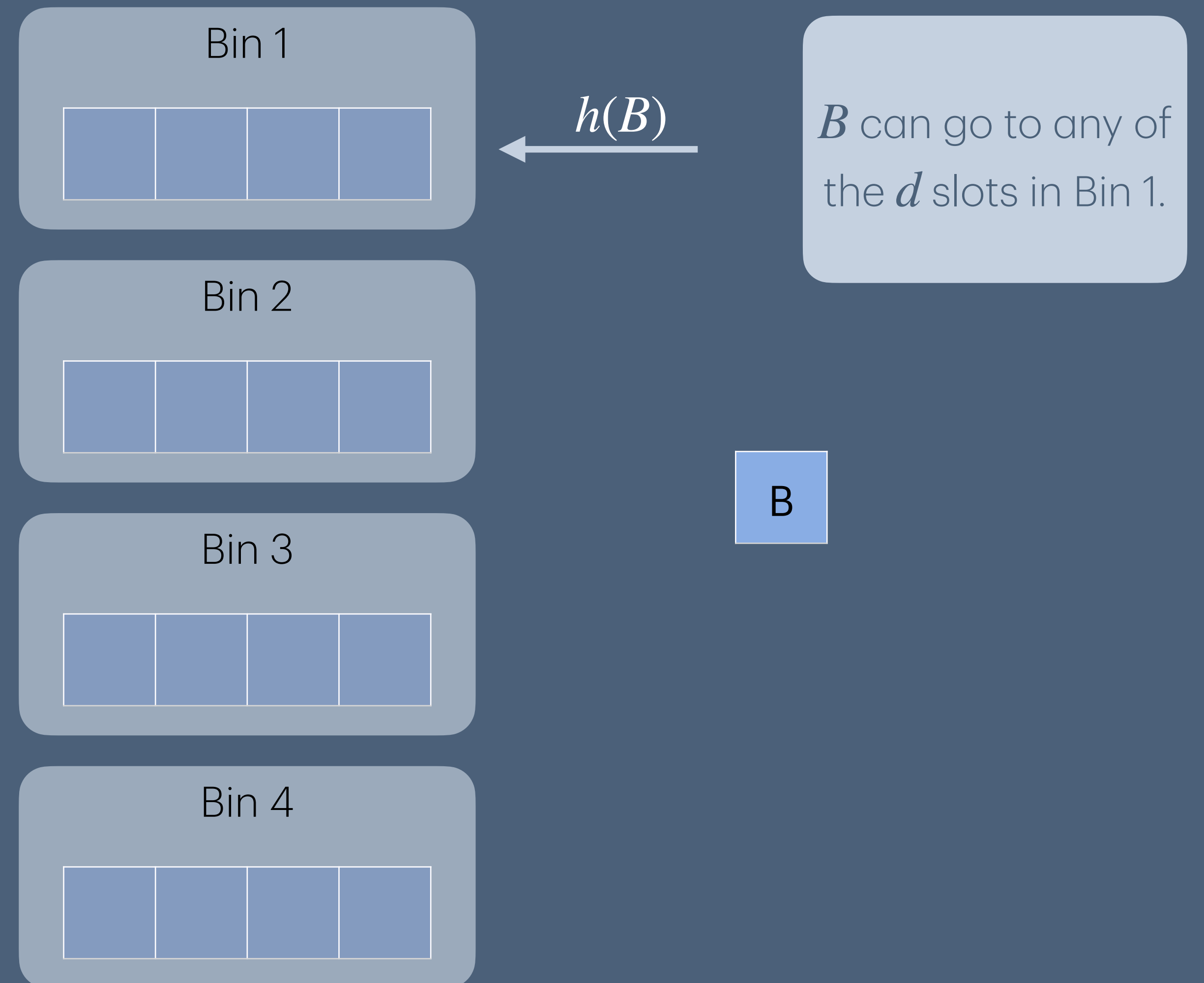
Question: Can we modify *d-LRU*
to perform as well as LRU?

Well-Studied Variant— *d*-bin-LRU

d-bin-LRU: $H(x)$ maps to a *bin* of size d . Pairwise disjoint bins, *LRU* eviction.

Thm [BDFT, SPAA'23]: *d*-bin-LRU is not $O(1)$ -competitive with $O(1)$ augmentation for $d \in o(\log n)$.

Culprit: Overused bins; they see far too many requests and evictions.

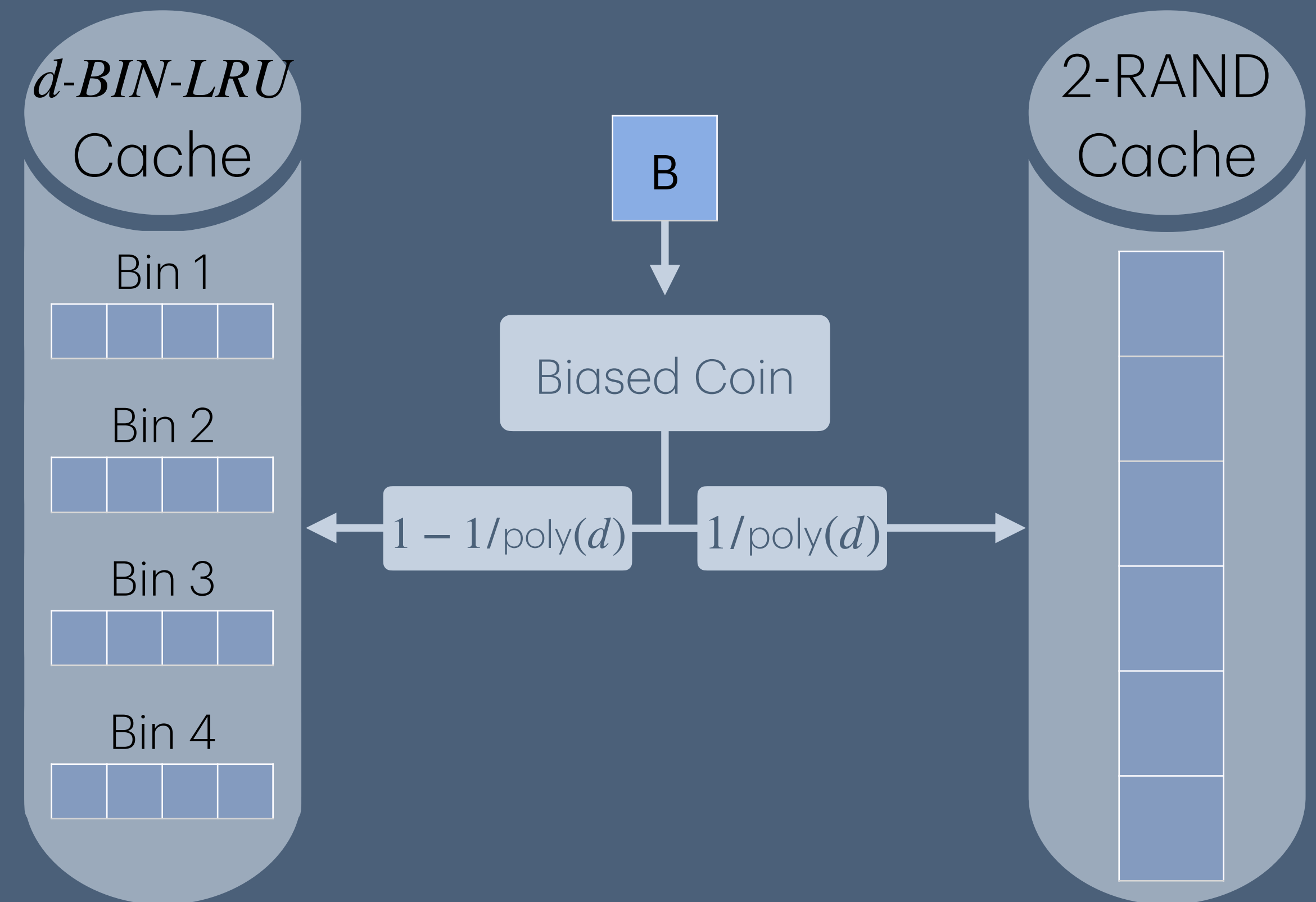


Small Amounts of Randomness Save Overused Bins.

HEAT-SINK-LRU: A n -sized d -bin-LRU cache with a $o(n)$ -sized *2-RAND* cache.

With probability $1/\text{poly}(d)$, the page enters *2-RAND* instead.

Overall, the associativity is $d + 2$.

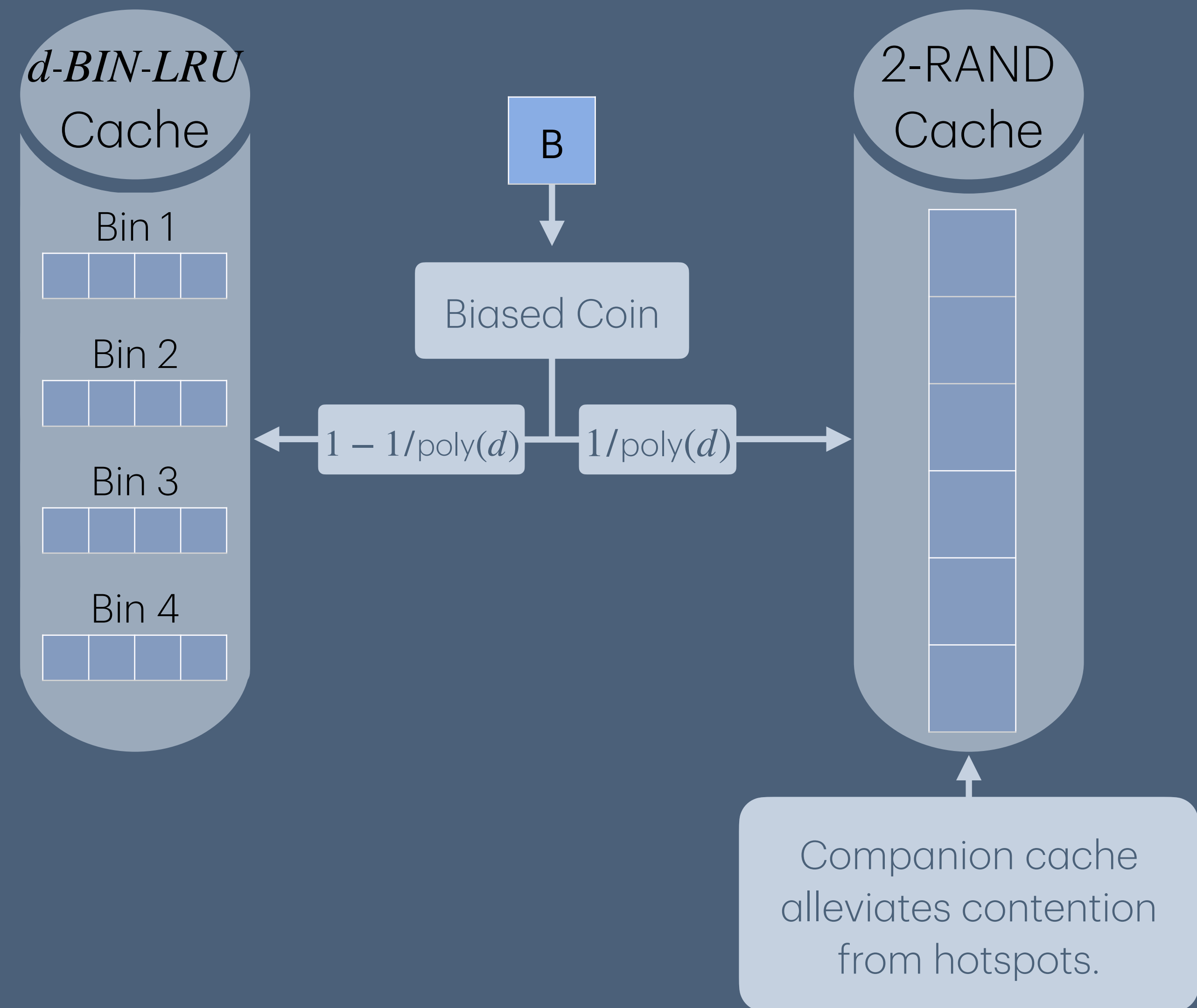


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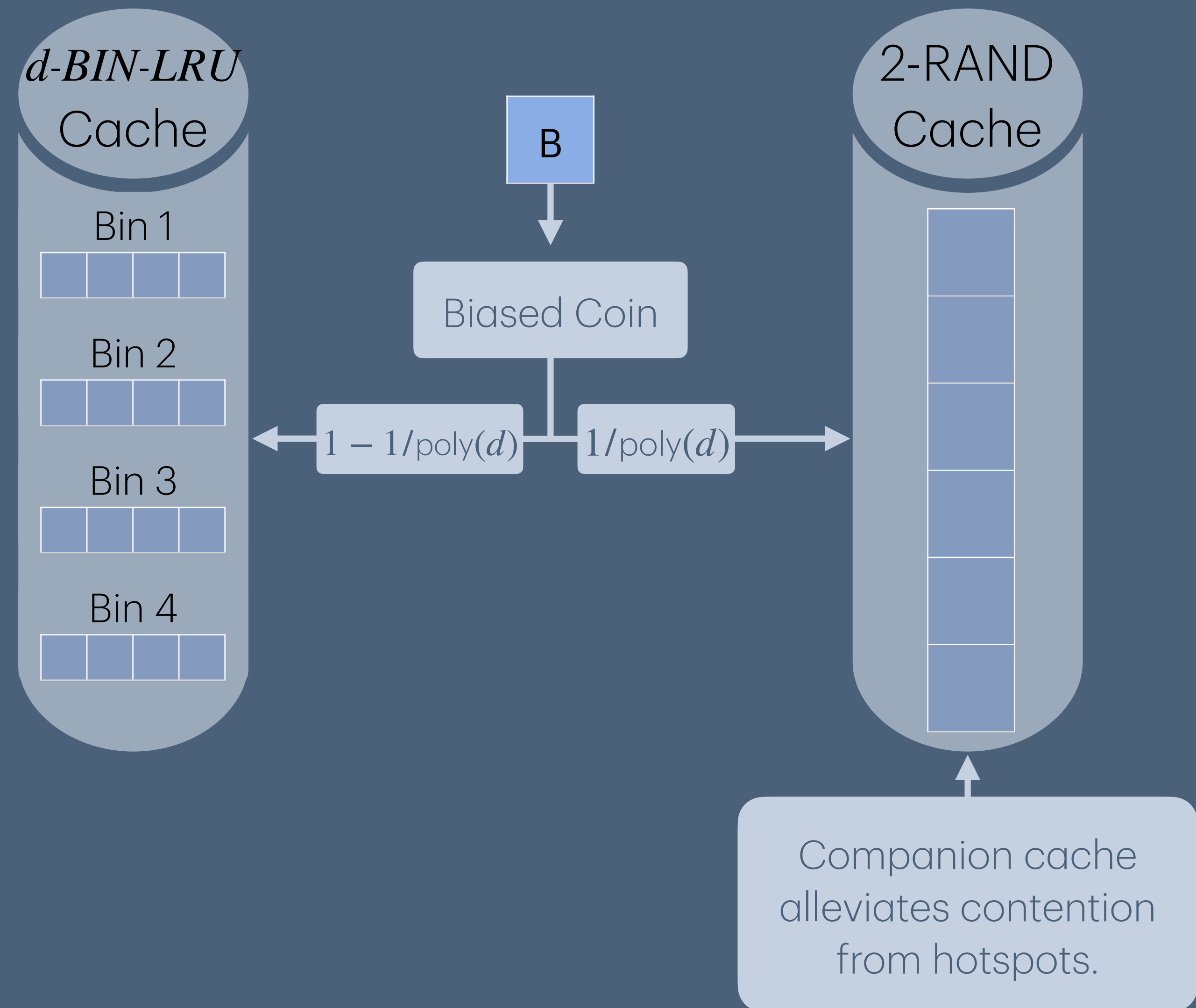
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See paper for details.



Question: *Can a low-associativity policy be competitive with ***OPT***?*

Answer: Yes;
Using the right algorithm.